USA: Pushing Back Fed Rate Hike on Lower Core PCE Inflation; Q2 GDP Revised Up to +2.2%; Trade Deficit Widens More Than Expected;

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BOTTOM LINE: The core PCE price index rose 0.25% in August, somewhat below expectations, and the year-over-year rate now stands at 3.01%, well below expectations. The downward revision to year-over-year core PCE growth resulting from the methodological changes was larger than we expected and mostly reflected a larger revision to the portfolio management component. Based on today's inflation report, we now expect 3.0% growth for core PCE on a Q4/Q4 basis, well below the median FOMC participant's forecast of 3.4%. Coupled with comments from New York Fed President John Williams yesterday, we now think that an October hike is unlikely. We are pushing back the second hike in our forecast to December, and we see a strong chance that the FOMC will ultimately conclude that additional rate hikes are unnecessary. Real GDP was revised up by 0.7pp to +2.2% in its third Q2 release (quarter-over-quarter annualized), largely reflecting upward revisions to consumption and investment growth. The annual revisions to the national accounts were released today. Real GDP growth was revised up by 0.4pp to +2.5% in 2026Q1 but revisions to prior years were modest. The savings rate was revised up significantly and stands at 4.1% in August. The goods trade deficit widened by more than expected in August. The details of the personal income and spending report were firmer than our previous assumptions, while the details of the Advance Economic Indicators report were on net weaker: we lowered our Q3 GDP tracking estimate by 0.1pp to +3.3% (quarter-over-quarter annualized).

US MAP:

Personal spending 0 (1, 0)

ADP 0 (3, 0)

KEY NUMBERS:

Core PCE price index +0.25% (MoM) for August vs. GS +0.27%, median forecast +0.3%, prior revised +0.13%

Core PCE price index +3.01% (YoY) for August vs. GS +3.17%, median forecast +3.3%, prior revised +2.98%

PCE price index +0.31% (MoM) for August vs. GS +0.33%, median forecast +0.3%, prior revised +0.05%

PCE price index +3.42% (YoY) for August vs. GS +3.58%, median forecast +3.7%, prior revised +3.36%

GDP (third estimate) +2.2% (qoq ar) for Q2 vs. GS +1.5%, median forecast +1.5%, prior revised +2.5%

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Personal consumption (third estimate) +3.8% (qoq ar) for Q2 vs. GS +3.4%, median forecast +3.4%, prior revised +0.7%

Personal income +0.2% for August vs. GS +0.5%, median forecast +0.5%, prior revised +0.3%

Personal spending +0.9% for August vs. GS +0.8%, median forecast +0.9%, prior revised +0.1%

Advance goods trade balance -$132.6bn for August vs. GS -$116.0bn, median forecast -$115.3bn, prior revised -$118.9bn

ADP employment +90k for September vs. GS +75k, median forecast +75k, prior revised +36k

MAIN POINTS:

1. The core PCE index increased +0.25% in August, somewhat below expectations, and the year-over-year rate increased 3.01%. We estimate that the Dallas Fed's trimmed mean PCE measure increased 0.16% month over month and 2.20% year over year.

2. Year-over-year core PCE was revised downward 0.36pp in July to 2.98%, primarily reflecting the methodological revisions to the calculation of the portfolio management (-0.23pp), computer accessories and software (-0.11pp), and legal services (+0.01pp) components of PCE. These downward revisions were larger than we expected, mostly reflecting a larger revision to the portfolio management component than we expected.

3. Today's inflation report, especially the downward revisions resulting from the methodological changes, imply that core PCE inflation is likely to substantially undershoot the median FOMC participant's forecast of 3.4% for 2026 on a Q4/Q4 basis. We now expect a 3.0% increase for core PCE in 2026 on a Q4/Q4 basis. Coupled with comments from New York Fed President John Williams yesterday that "with the policy action we took at our September meeting, there is no need for urgency, and we have time to gather more information," we now think that an October hike is unlikely. We are pushing back the second hike in our forecast to December, and we see a strong chance that the FOMC will ultimately conclude that additional rate hikes are unnecessary.

4. Real GDP was revised up by 0.7pp to +2.2% in Q2 (quarter-over-quarter annualized), largely reflecting upward revisions to consumption (+0.4pp to +3.8%) and investment (+1.9pp to +4.6%) growth. The strength in consumer spending in Q2 likely continued to reflect the boost to consumer cashflow from larger-than-usual tax refunds, but we continue to expect consumer spending growth to slow somewhat in 2026Q4 as the fiscal support fades and the headwind from higher energy prices remains. Stronger investment growth reflected an upward revision to private fixed investment (+0.5pp to +9.0%), driven by data center investment in the nonresidential structures category (+1.9pp to +0.1%), and an upward revision to the contribution from inventories (+0.2pp to -0.5pp).

5. Reflecting today's annual revisions to the national accounts, real GDP was revised up by 0.4pp to +2.5% in Q1 (quarter-over-quarter annualized), largely reflecting upward revisions to the contributions from consumption and exports. Revisions for prior years were modest: real GDP growth on a Q4/Q4 basis was unrevised for 2025 (+2.0%) and 2023 (+3.4%), and slightly revised down for 2024 (-0.2pp to +2.4%) and up for 2022

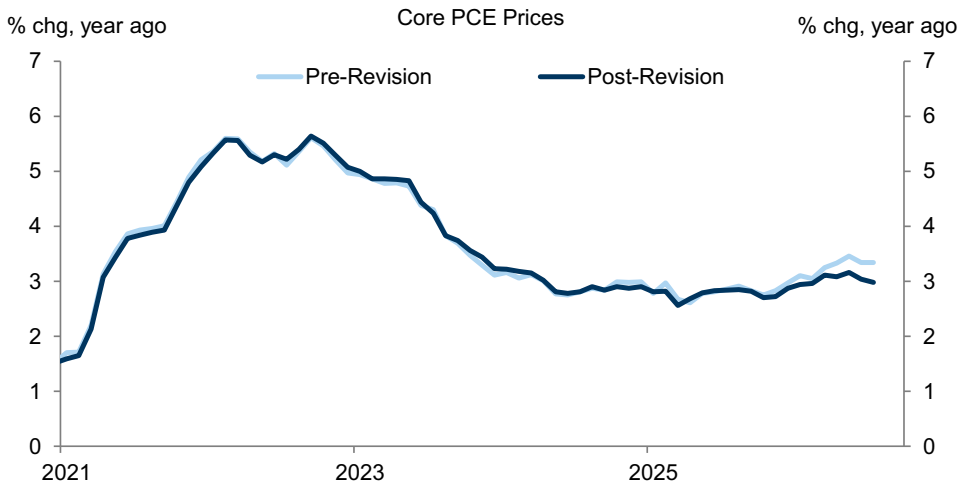
(+0.1pp to +1.3%).

6. Personal income increased 0.2% in August, below expectations. Year-over-year disposable personal income growth reached 4.8% in August. Personal spending rose 0.9% in August, in line with consensus expectations. The savings rate decreased 0.5pp to 4.1% in August, after a large 1.6pp upward revision to the savings rate last month driven by an upward revision to the level of disposable personal income (+1.2pp contribution) and a downward revision to personal spending as a share of DPI (+0.4pp).

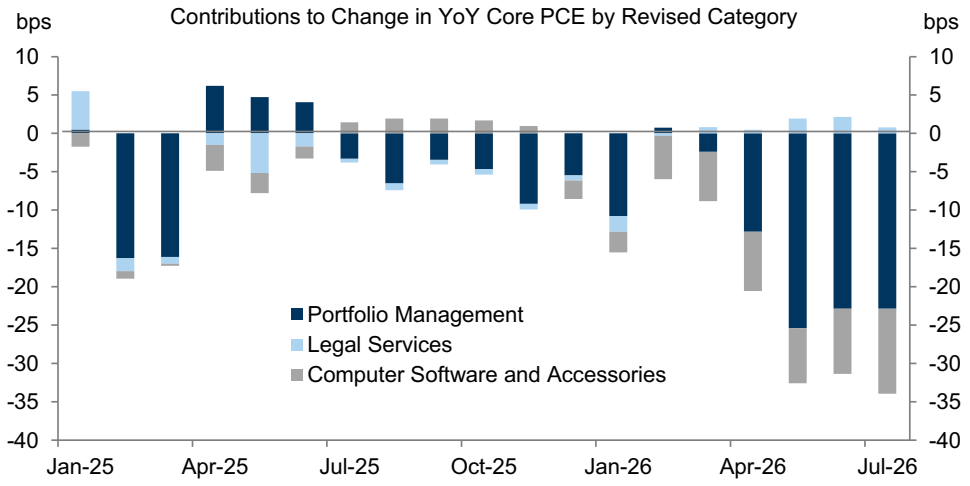
7. According to the Advance Economic Indicators Report, the goods trade deficit widened by $13.7bn to $132.6bn (seasonally adjusted) in August, well above expectations. Goods exports increased by $3.7bn to $203.4bn, largely driven by an increase in industrial supplies exports. Goods imports increased by $17.4bn to $336.1bn, driven by a large increase in industrial supplies and capital goods imports.

8. According to the ADP report, private sector employment rose by 90k in September (month-over-month, seasonally adjusted), slightly above consensus expectations. Employment in the services sector rose 59k, primarily reflecting a 55k increase in education and health services and a 22k increase in leisure and hospitality employment that were partly offset by a 16k decrease in financial activities employment. Employment in goods-producing industries increased by 31k, driven by a 17k increase in manufacturing and a 15k increase in construction employment. ADP's new median annual base pay measure, which better tracks other measures of wage growth than its gross pay measure, increased 3.0% for job stayers, unchanged from the August pace, and 4.8% for job changers, 0.1pp above the August pace. We left our forecast for August nonfarm payroll growth unchanged at +80k (vs. +90k consensus) ahead of Friday's release.

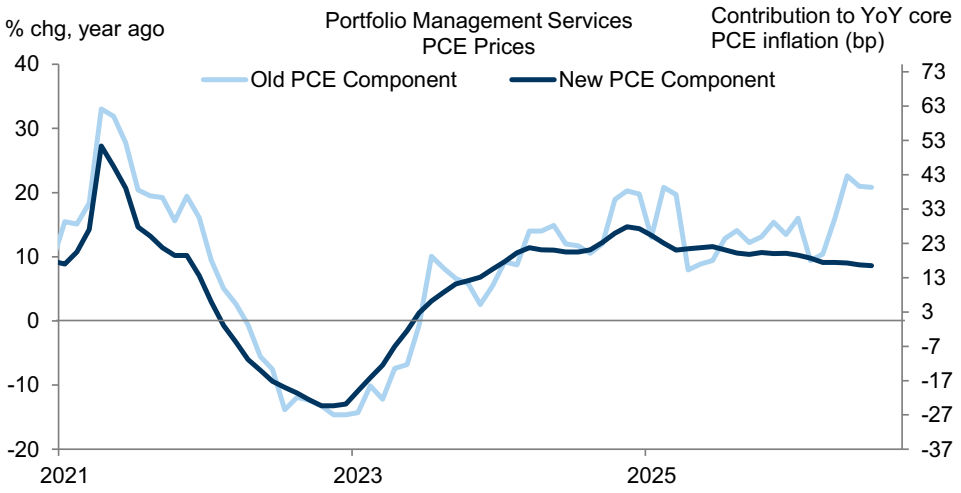
9. The details of the personal income and spending report were firmer than our previous GDP tracking assumptions. While the trade-related details of the Advance Economic Indicators report were much weaker than our previous assumptions, suggesting notably stronger imports growth in Q3 than we previously assumed, the inventories details were significantly stronger. On net, we lowered our Q3 GDP tracking estimate by 0.1pp to +3.3% (quarter-over-quarter annualized).



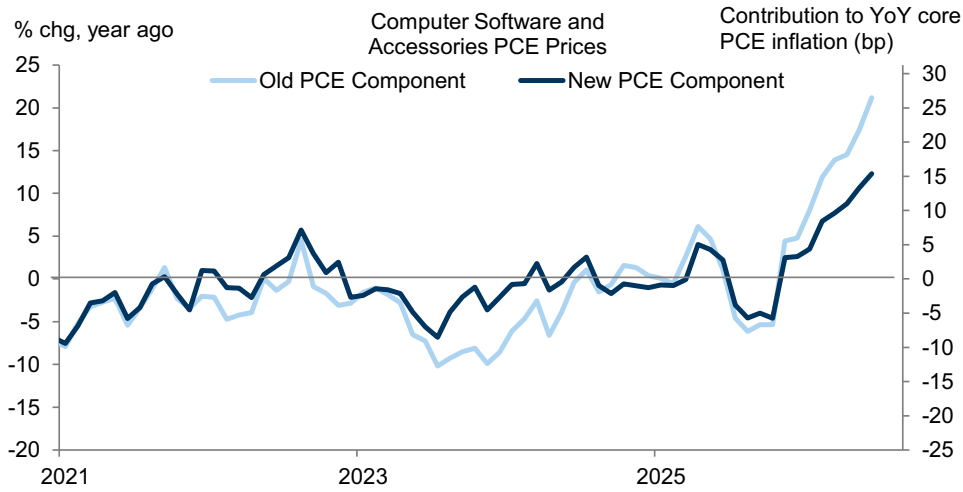
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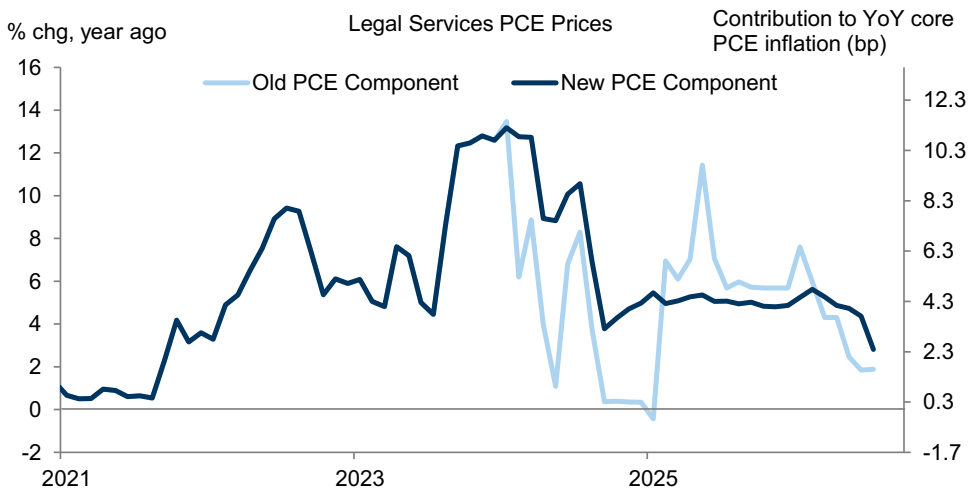
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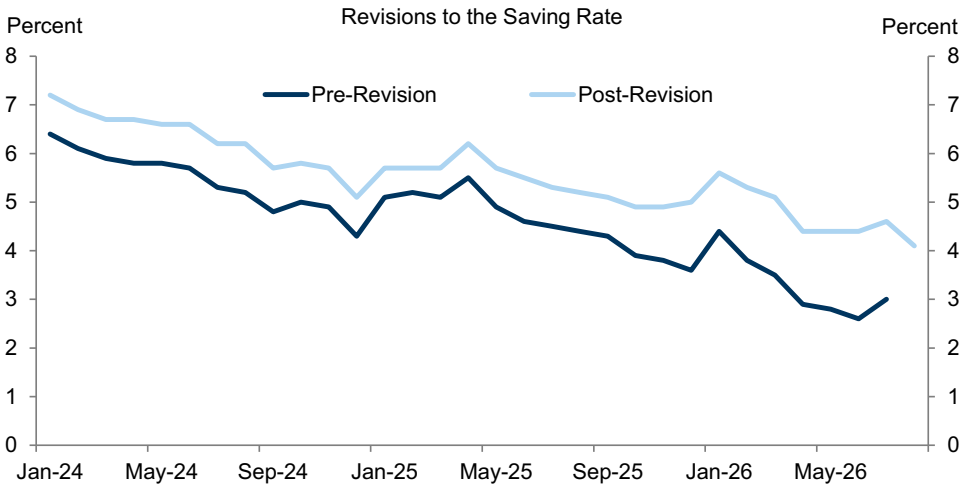
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US Daily: Why Has the US Economy Been So Resilient to Higher Rates and Higher Oil Prices?

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- We estimate that the economy grew at a 3.4% pace in Q3 and forecast 2.2% growth in 2026 overall on a Q4/Q4 basis, close to our estimate of potential GDP growth and only 0.3pp below our start-of-the-year forecast before the large increases in interest rates and oil prices. Why, we are often asked, has the US economy proven so resilient to these shocks?
- While interest rates have risen substantially this year, our financial conditions index framework implies that the economic impact should be nearly offset by the effects of higher equity prices, narrower credit spreads, and a weaker dollar, resulting in only a 0.1pp net hit to 2026 GDP growth. While the net impact of this year's market moves on overall GDP is modest, the effects on some parts of the economy—especially the hit to housing—are more meaningful.
- We estimate that the increase in oil and refined product prices this year, together with the path our commodities strategists expect through year-end, will subtract 0.3pp from 2026 growth. The drag comes from consumption as higher energy prices erode real purchasing power. While US energy producers were reluctant to boost investment when oil prices first rose, the increase in the rig count index in recent months points to some offsetting boost to GDP in Q3 and Q4.
- Our analysis suggests that slightly tighter financial conditions and higher oil prices should together be roughly a 0.4pp drag on GDP growth. The downgrade we have made to our forecast for the year, in part based on incoming data, is only slightly smaller than that, with the slight outperformance coming largely from consumer spending. This suggests that the economy's resilience is less puzzling than it might appear at first.

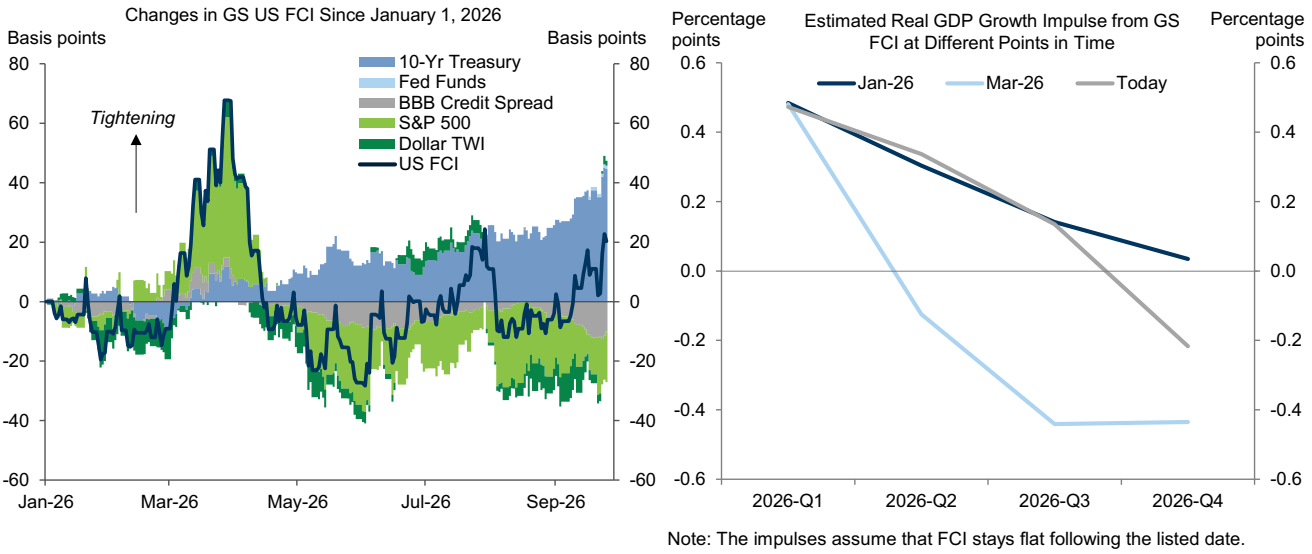
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Why Has the US Economy Been So Resilient to Higher Rates and Higher Oil Prices?

We now estimate that the economy grew at a 3.4% pace in Q3 and forecast 2.2% growth in 2026 overall on a Q4/Q4 basis, close to our 2.3% estimate of potential GDP growth and only 0.3pp below our start-of-the-year forecast when we did not expect large increases in interest rates and oil prices. In today's *Daily*, we ask why the US economy has proven so resilient to these surprises.

Part of the answer is that financial conditions have changed little on net since January, as the substantial rise in interest rates has been nearly offset by the effects of higher equity prices, narrower credit spreads, and a weaker dollar (Exhibit 1, left). Our estimated GDP growth impulse from the FCI is therefore little changed from the start of the year (Exhibit 1, right), implying only a 0.1pp hit to 2026 Q4/Q4 GDP growth.

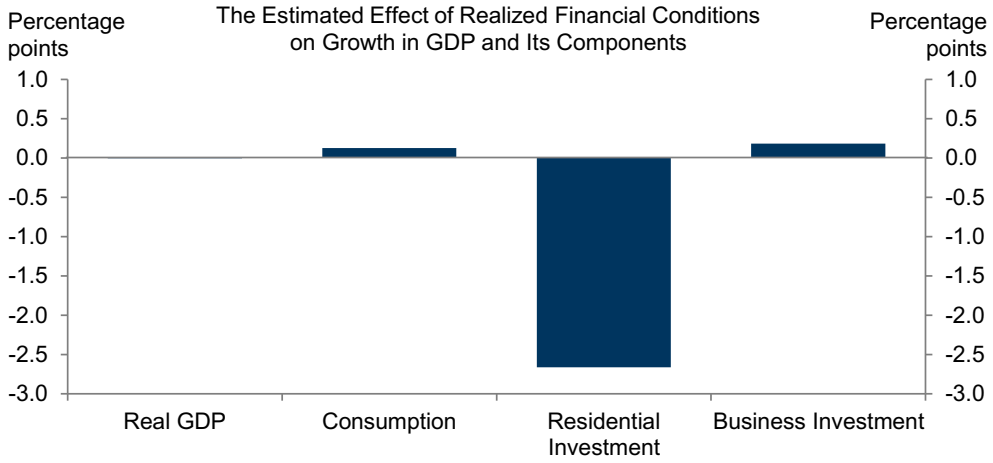
Exhibit 1: Our Overall Financial Conditions Index and the Implied Impulse to GDP Growth Have Changed Little Since the Start of the Year, as Higher Interest Rates Have Been Offset by Tighter Credit Spreads, Higher Equity Prices, and a Weaker Dollar



Source: Goldman Sachs Global Investment Research

While the net impact of this year's market moves on overall GDP is modest, the effects on some components of GDP are more meaningful. We use the Fed's macroeconomic model FRB/US to estimate how the particular combination of market moves this year should affect each component of GDP. We find a meaningful hit from higher rates to residential investment, offset by slight boosts to consumption and business investment (Exhibit 2).

Exhibit 2: This Year's Combination of Market Moves Should Have Only a Modest Effect on Overall GDP, but Will Impose a Larger Drag on Residential Investment, Which Is More Sensitive to Higher Interest Rates



Note: Estimates are calculated using FRB/US simulations comparing Q4/Q4 growth for each component under this year's changes in market conditions versus a counterfactual where conditions were unchanged from the start of the year.

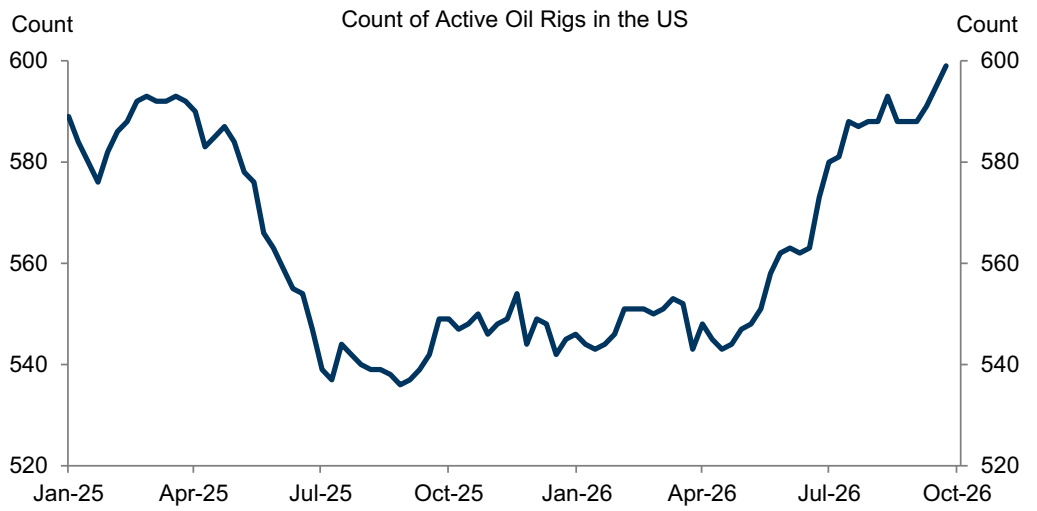
Source: Goldman Sachs Global Investment Research, Federal Reserve Board

What about the effect of higher oil and refined products prices? Our rule of thumb implies that a sustained $10/barrel increase in oil prices lowers GDP growth by just over 0.1pp through the consumption channel as higher energy costs erode real purchasing power. Applied to the path oil prices have taken this year, the unusual widening in refined product spreads, and our commodities strategists' forecasts for prices through year-end, the rule of thumb implies a 0.4pp drag on GDP growth.

While US oil producers were initially reluctant to respond to higher prices by increasing capex because of the uncertainty of the war with Iran and the outlook for oil prices, the increase in the rig count index in Q3 (Exhibit 3) points to a pickup in energy capex more recently. We estimate that the consumption drag and capex boost will net out to a 0.3pp hit to GDP growth from the oil shock. Two alternative approaches, FRB/US simulations and a model estimated using exogenous oil shocks, deliver similar estimates on average.[1]

[1] We use the series of exogenous oil shocks from Känzig (2021), "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements" to estimate the effects of the current oil shock on GDP growth.

Exhibit 3: The Recent Uptick in Active Oil Rigs Suggests an Offsetting Boost to Business Investment from Energy Capex Through the Second Half of the Year



Source: Goldman Sachs Global Investment Research, Baker Hughes

Our analysis suggests that slightly tighter financial conditions and higher oil prices should together be roughly a 0.4pp drag on GDP growth. The downgrade we have made to our forecast for the year, in part based on incoming data, is only slightly smaller than that, suggesting that the puzzle of economic resilience is smaller than it might appear at first.

The economy's slight outperformance relative to what we might have expected largely comes from consumer spending. At the start of the year, we forecasted consumption growth of 2.4% in 2026 Q4/Q4. Under the market and commodity price moves that actually materialized, our oil shock model implies a 0.6pp hit to that forecast, partially offset by a 0.2pp boost from higher equity prices in our wealth effects model, netting to an expected 2.0% growth in consumption this year. Against that, we now estimate—in part based on the surprisingly solid incoming data in recent months—that consumption will grow 2.2%, a modest upside surprise of 0.2pp.

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Disclosure Appendix

Reg AC

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24 September 2026 | 7:04PM EDT

US EQUITY VIEWS

AI capex, required revenues, and what's priced in the AI trade

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Outlook for AI capex: The largest US AI hyperscalers are on track to spend $800 billion in capex in 2026 and consensus expects $1.1 trillion in spending in 2027. Our baseline forecast is that capex spending will increase in 2027 and surpass consensus estimates but the rate of both capex growth and upside surprises will diminish relative to recent quarters.

What capex signals about required AI revenues: Based on average annual capex in 2026 and 2027, our equity analysts estimate that the hyperscalers need to generate annual AI revenues of roughly $300 billion in the next few years to break even on their investments. Hyperscaler cloud revenues have accelerated sharply this year, annualizing roughly $70 billion above the pre-AI trend in Q2 2026, and announced revenue backlogs exceed $1.5 trillion.

What required AI revenues signal about enterprise and consumer AI spending: We estimate that AI users would need to spend roughly $1 trillion annually on AI applications in order for the hyperscalers to generate solid returns on investment and the application layer to generate strong profit margins on their compute expenses. This compares with roughly $1.5 trillion in global software spending in 2026 and $1.8 trillion in potential annual US labor productivity gains that our economists estimate will eventually accrue to capital. However, not all of these productivity gains will necessarily accrue to the infrastructure and application layers. AI-related revenues and capex needs would be substantially larger accounting for physical AI or the potential emergence of AGI.

What the AI infrastructure trade is pricing: Many market participants are skeptical about the durability of AI infrastructure stock earnings, and that view is at least partially reflected in market pricing. At current share prices, earnings among some semiconductor stocks would have to decline substantially for the groups to trade in line with their historical average P/E multiples. The hyperscalers also trade at low earnings multiples, and the combination of increasing signs of ROI and the likelihood of decelerating capex growth should be supportive for these stocks.

The AI trade outside of the infrastructure layer: We expect the application layer will remain a source of return dispersion, as reflected by the recent outperformance of data infrastructure and cybersecurity stocks and underperformance of "consumer inertia" stocks. Enterprise adoption is still in early stages but the recent acceleration in enterprise AI spending suggests the impact of AI on corporate earnings should become increasingly visible in coming quarters. Nonetheless, current market-implied long-term growth expectations are only slightly above historical averages.

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An update on AI and US equities

Both stocks and investor views relating to the AI trade have been exceptionally volatile during the last few months. Most of the volatility has been concentrated in the AI infrastructure complex, but the latest earnings season also drove sharp moves in the hyperscalers and among software stocks.

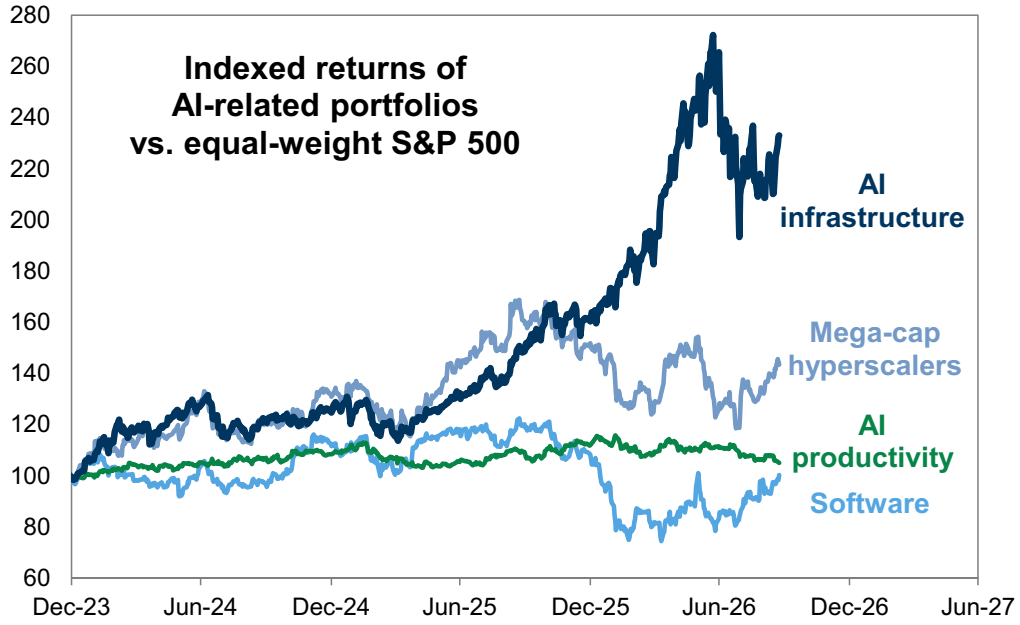
Below we address three of the most common AI-related questions we have received from clients in recent weeks:

1. What is the outlook for AI capex?

2. What revenues are required to generate a sufficient return on recent AI investments?

3. What outlook for AI is the equity market pricing?

Exhibit 1: Performance of the phases of the AI trade



"AI infrastructure" = GSCBAIP2, "Mega-cap hyperscalers" = AMZN, GOOGL, META, MSFT, ORCL, "Software" = IGV, "AI productivity" = GSXUPROD. Baskets developed by GBM.

Source: Goldman Sachs Global Investment Research

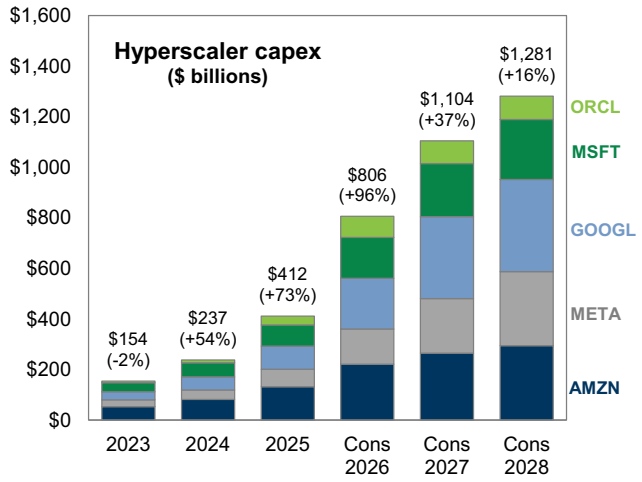
1. What is the outlook for AI capex?

The largest US hyperscalers are on track to spend $800 billion in capex in 2026 and consensus estimates imply they will spend $1.1 trillion in 2027. Consensus 2027 capex estimates have been revised higher by $175 billion since the start of the Q2 earnings season. Commentary from the hyperscalers during earnings season described a supply-constrained environment and large revenue backlogs, and these factors are driving the companies to tap both debt and equity markets to support additional investment spending. Based on consensus estimates, capex in 2027 is on track to reach a

larger share of GDP than any technological investment cycle since the railroad build-out in the late 1800's.

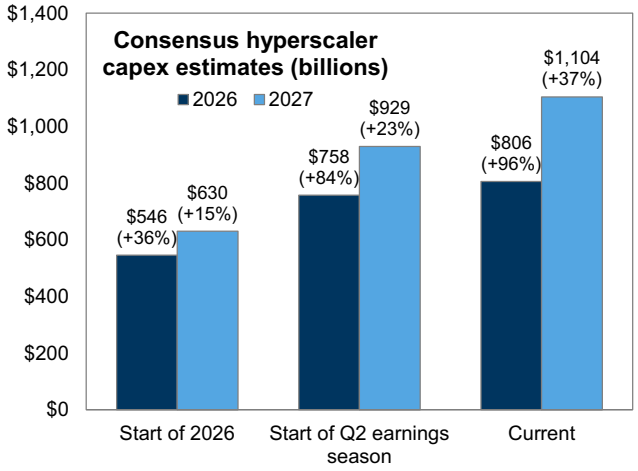
Our baseline forecast is that hyperscaler capex spending will increase substantially in 2027 but that the rate of both capex growth and upside surprises to consensus estimates will diminish relative to recent quarters. GS equity analysts estimate that hyperscaler capex will register $1.2 trillion in 2027 and $1.4 trillion in 2028. Their estimates are $130 billion and $100 billion above the consensus estimates, respectively, but nonetheless imply a deceleration in capex growth from nearly 100% in 2026 to 54% in 2027 and 12% in 2028.

Exhibit 2: Consensus hyperscaler capex estimates



Source: FactSet, Goldman Sachs Global Investment Research

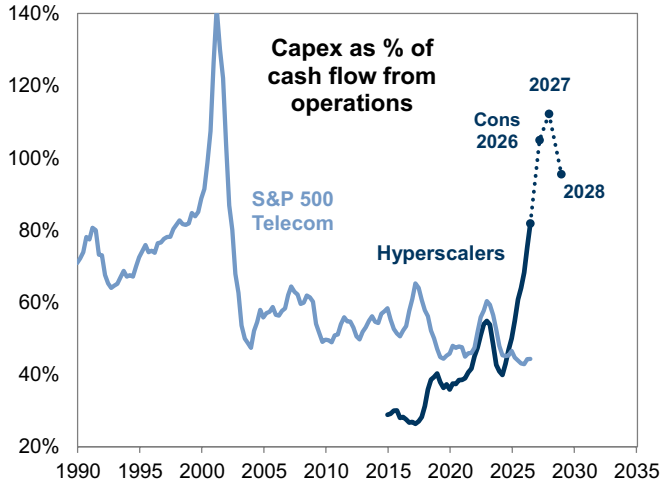
Exhibit 3: Capex estimates have been revised higher



Source: FactSet, Goldman Sachs Global Investment Research

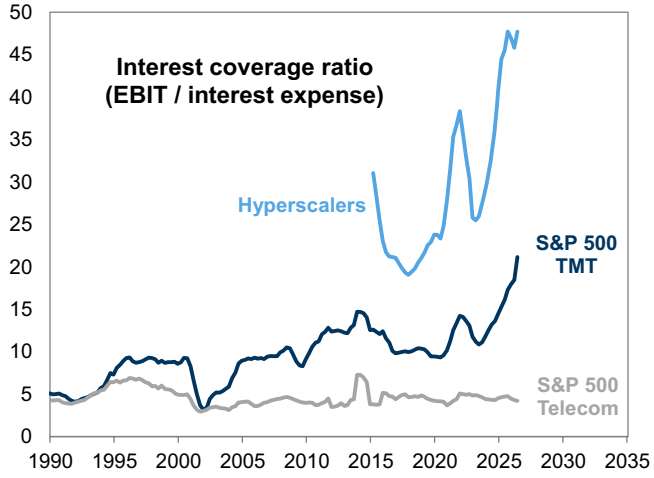
The need for external financing and potential supply bottlenecks will likely constrain the rate of capex growth next year. Hyperscaler capex spending now exceeds cash flow from operations, which means large further growth will increasingly need funding via debt and equity issuance. While hyperscalers have added leverage, their balance sheets do not appear to be a binding constraint and their interest coverage ratios remain extremely healthy. Our credit strategists instead point to concentration in the investment grade market as the more relevant bottleneck, which will require additional funding from private credit and equity issuance. Potential government restrictions on data center construction or an intentional slowdown in frontier model roll-out could also act as potential constraints on spending, as could shortages of labor, power, memory, or other inputs. Constraints on capex represent a risk to the near-term earnings of AI infrastructure beneficiaries, but could help ease investor concerns about overbuilding.

Exhibit 4: Hyperscaler capex expected to surpass cash flows from operations this year



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 5: Interest coverage ratios remain extremely healthy versus history

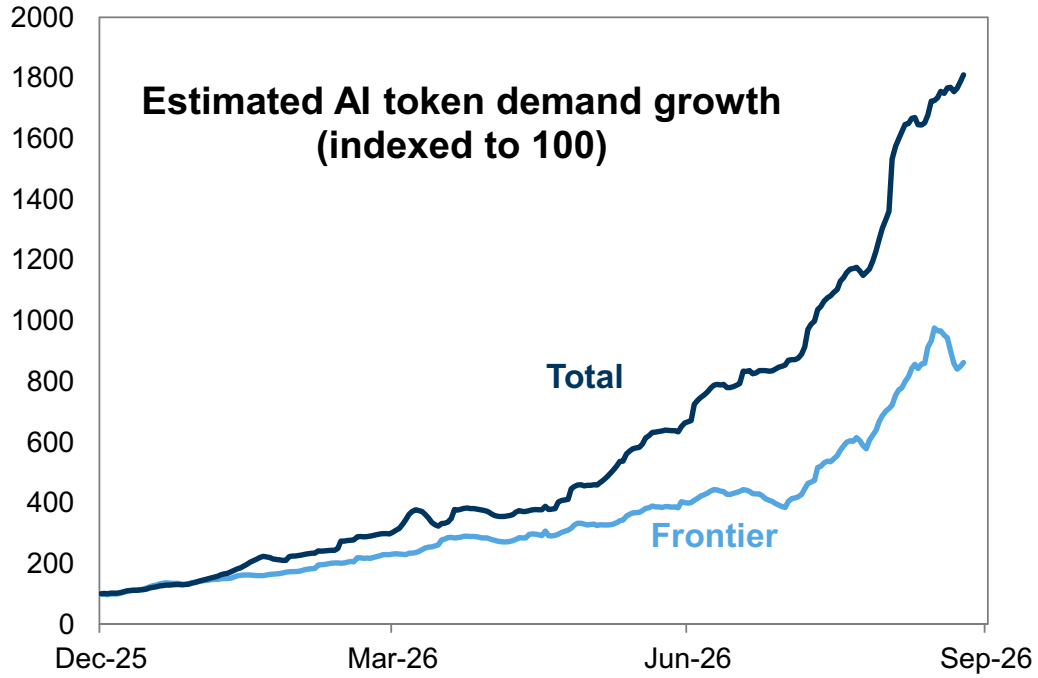


Source: Compustat, Goldman Sachs Global Investment Research

Going forward, token demand growth will need to outpace declining token prices to support continued growth in investment spending. Frontier models are currently a key source of demand for hyperscaler compute. However, the rise of competitive open-source models has contributed to a decline in average token prices. Measures of frontier token demand slowed in July, simultaneous with sell-offs of many AI infrastructure stocks, but accelerated again in August.

For most AI infrastructure companies, decelerating AI investment spending suggests decelerating EPS growth. AI infrastructure companies, and semis in particular, have been the clearest beneficiaries of the capex surge. Continued upward revisions to consensus 2027 capex estimates should support continued positive revisions to consensus 2027 EPS estimates for many AI infrastructure stocks, but the magnitude of revisions will likely diminish relative to what occurred in H1 2026.

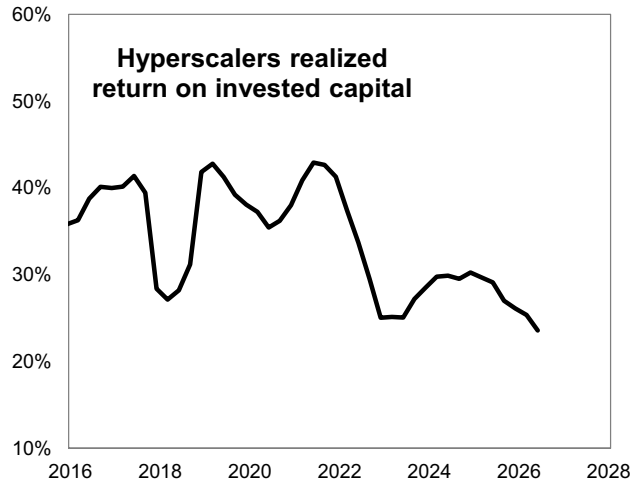
Exhibit 6: Estimated token demand growth
lines represent equal-weighted averages of growth implied by Anera and OpenRouter data



Source: Anera, OpenRouter, Goldman Sachs Global Investment Research

2. What revenues are required to generate a return on AI investments?
Our equity analysts estimate that the hyperscalers would need to generate annual AI revenues of roughly $300 billion in the next few years to break even on their average annual capex investments in 2026 and 2027. These estimates reflect their expectations for average annual AI capex in 2026 and 2027 for the US hyperscalers, including AMZN, GOOGL, META, MSFT, ORCL, and SPCX. The breakeven revenues required for this capex represent the sum of annual depreciation and operating expenses. Exhibit 8 provides an estimated sensitivity of required AI revenues the hyperscalers must generate in different return on invested capital ("ROIC") scenarios. These required revenue estimates are anchored on average annual capex in 2026 and 2027 and would argue for even larger estimates based on cumulative capex.

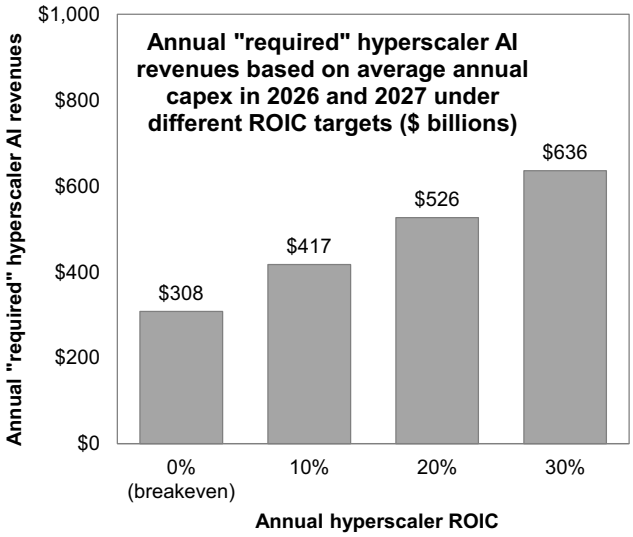
Exhibit 7: Realized hyperscaler return on invested capital has declined from 40% to 25%



Hyperscalers include AMZN, GOOGL, META, MSFT, ORCL. ROIC = (EBIT less taxes) / (equity plus debt less cash).

Source: Compustat, Goldman Sachs Global Investment Research

Exhibit 8: Sensitivity of annual "required" hyperscaler AI revenues to ROIC assumption

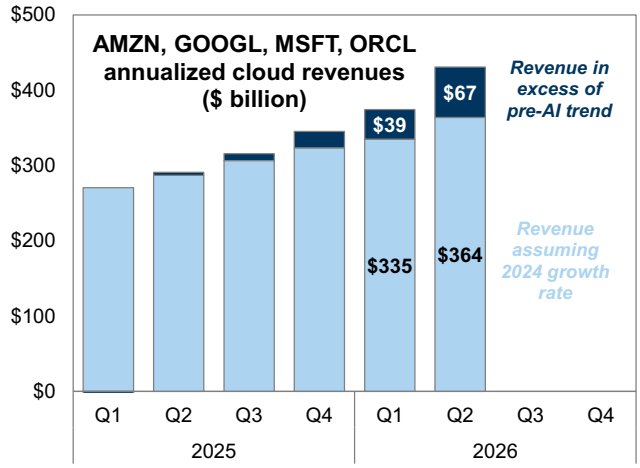


Hyperscalers include AMZN, GOOGL, META, MSFT, ORCL, and SPCX. "Required" AI revenues equal the sum of the estimated annual depreciation, annual operating expenses, and return on invested capital using gross capex, based on GS estimate of average annual AI capex in 2026 and 2027. See Sizing the AI Economy Needed to Justify ROIC on Hyperscaler AI Capex for more details.

Source: Goldman Sachs Global Investment Research

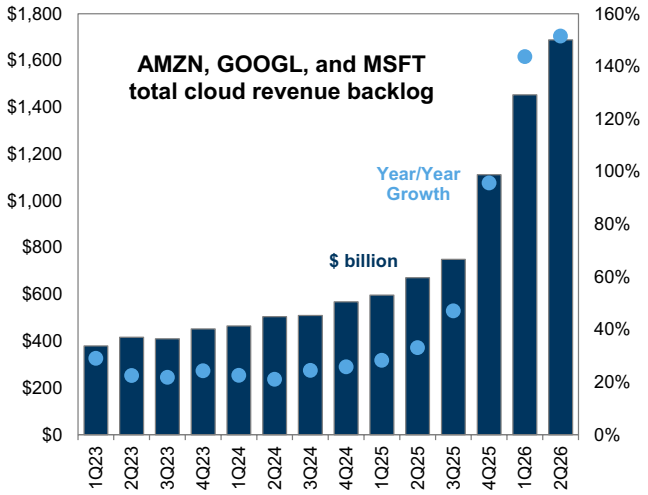
The hyperscalers' current estimated AI revenues remain below the amount that would allow them to break even on their capex, but those revenues are growing quickly and revenue backlogs are sizable. The hyperscalers have started to monetize their AI investments. Year/year cloud revenue growth for AMZN, GOOGL, MSFT, and ORCL has accelerated from 25% in 2024 to 48% in Q2 2026. In dollar terms, annualized cloud revenues in excess of their 2024 growth rate registered nearly $70 billion in Q2 and have been roughly doubling during the past few quarters. The three major US public cloud companies (AMZN, GOOGL, MSFT) have also announced a combined revenue backlog of $1.7 trillion, which provides line of sight into these companies' ability to meet required revenue targets.

Exhibit 9: Sizing cloud revenues in excess of 2024 realized growth rate



Source: Goldman Sachs Global Investment Research

Exhibit 10: The large public US cloud providers have announced sizable backlogs



Source: Goldman Sachs Global Investment Research

The magnitude of capex in 2026 and 2027 points to even larger "required" annual revenues in the next few years for the "application layer" of companies buying much of that compute from the hyperscalers. These revenues reflect the additional costs and profit margins of application companies on top of the "required" hyperscaler revenues selling compute. EBIT margins for large-cap software companies have averaged 40% during the past five years. We utilize EBIT margins rather than gross margins to include the costs of both training and inference, and assume additional expenses alongside compute. The range of potential required revenues is wide depending on which companies constitute the application layer and how their costs are structured.

Exhibit 11 provides a range of potential "required" revenues based on hyperscaler ROIC and application layer EBIT margins. For example, the application layer would have to generate over $1 trillion in annual revenues if those companies had a 30% EBIT margin and the hyperscalers generated a 10%-20% return on invested capital.

The "required" application layer revenues that would justify current capex are large but potentially achievable. For example, estimates of global advertising and software spending each equal roughly $1.5 trillion in 2026. Our economists estimate that total annual AI US labor productivity gains that will potentially accrue to capital total roughly $1.8 trillion. However, not all of these gains will necessarily accrue to the infrastructure and application layer companies. In addition, another year of strong capex growth in 2028 would raise the hurdle further. AI-related revenues and capex needs would be substantially larger accounting for physical AI or the potential emergence of AGI.

Exhibit 11: Sizing "required" revenues for the application layer

Annual "required" application layer revenues based on cumulative AI capex in 2026 and 2027 ($ billions)

Hyperscaler ROIC	Application layer EBIT margin			
	10%	20%	30%	40%
0%	$425	$475	$525	$625
10%	725	825	925	1,050
20%	1,050	1,150	1,300	1,500
30%	1,350	1,500	1,675	1,925

Based on GS equity analysts' estimates for cumulative AI capex for the hyperscalers in 2026 and 2027. "Required" application revenue assumes compute represents 75% of application layer operating expenses. EBIT margin not applied to META required revenues.

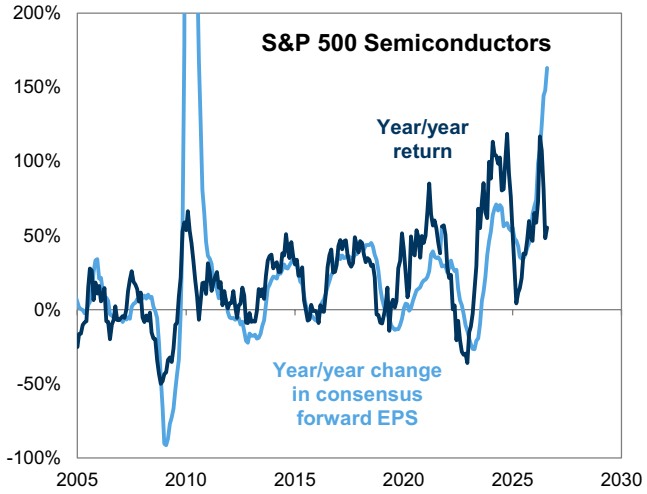
Source: Goldman Sachs Global Investment Research

3. What outlook for AI is the equity market pricing?

While market participants express skepticism about the durability of AI infrastructure stocks' earnings growth, that view is at least partially priced. The median AI infrastructure stock has de-rated from 32x consensus forward earnings in April 2026 to 22x today. In addition to tracking standard P/E multiples, comparing recent changes in market cap to recent changes in earnings can signal how long the market expects current earnings strength to persist. Among various AI infrastructure baskets, that ratio remains most elevated among optical networking, electrical grid, and liquid cooling stocks while expectations appear most conservative among semiconductors and utilities.

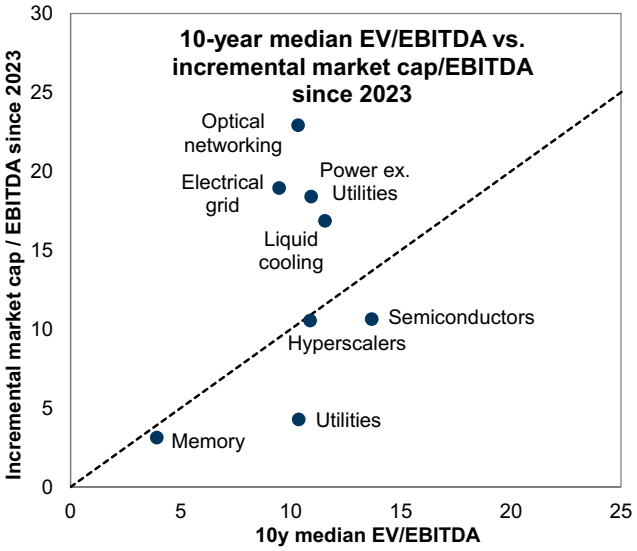
Semiconductor valuations appear to embed particularly skeptical views regarding the sustainability of current earnings strength. Valuations have declined sharply among Semis stocks, which is typical among cyclical companies with strong near-term growth but uncertain medium-term growth. We believe the investment opportunity is more compelling in fabless chip stocks than in memory stocks. Our equity analysts are Buy-rated on NVDA and AVGO but Neutral-rated on MU.

Exhibit 12: Semiconductor returns have decoupled from earnings



Source: FactSet, Goldman Sachs Global Investment Research

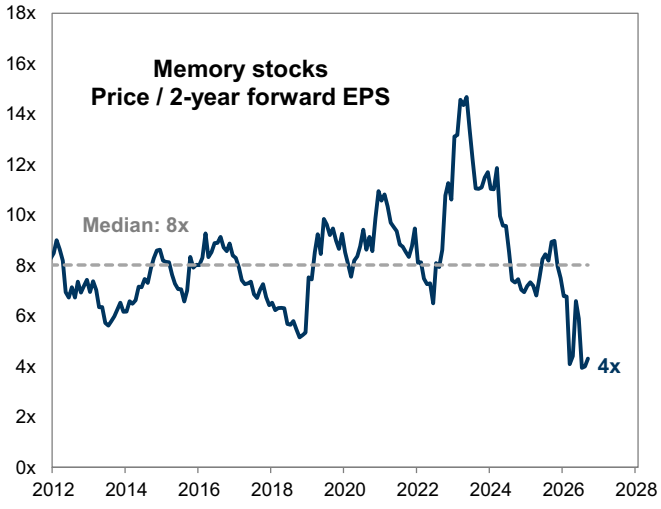
Exhibit 13: Expectations for earnings durability appear more conservative among Semis and Utilities



Source: FactSet, Goldman Sachs Global Investment Research

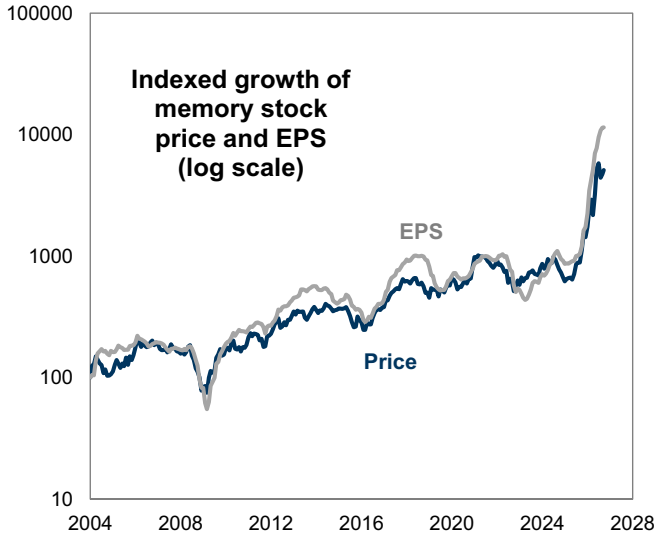
Memory stock prices imply earnings will decline in coming years, but the degree of pessimism does not appear particularly extreme relative to past cycles. Memory stocks currently trade at a 2-year forward P/E of 4x, half their 15-year average, but Memory firms currently enjoy gross margins of roughly 80%, more than double their historical average. To bring the current P/E of memory stocks to its 15-year average, earnings in coming years would need to decline by roughly 50%. While this decline is large in absolute terms, memory is a cyclical industry and has experienced larger EPS declines, including outright negative earnings, during previous cycles. This pricing will prove too pessimistic if 2028 forecasts are realized, but many investors have expressed skepticism about rotating back into higher beta pockets of the infrastructure trade as capex growth begins to normalize.

Exhibit 14: P/E of memory stocks
Memory represents US dollar-weighted combination of MU, SK Hynix, and Samsung



Source: FactSet, Goldman Sachs Global Investment Research

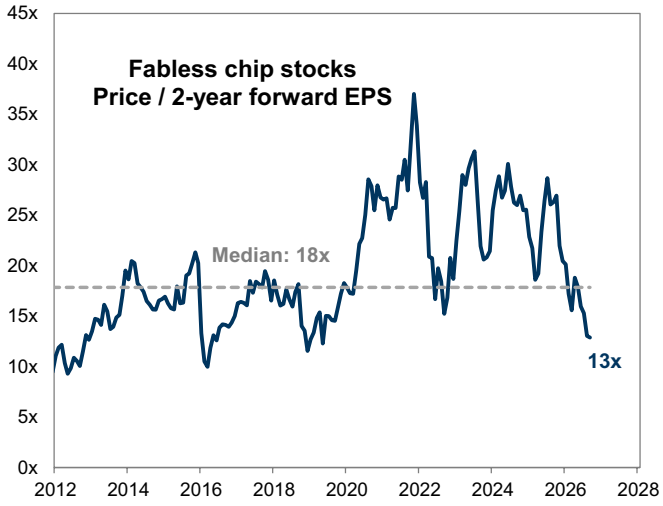
Exhibit 15: Price vs. EPS of memory stocks
Memory represents US dollar-weighted combination of Micron, SK Hynix, and Samsung; gray reflects consensus estimates



Source: FactSet, Goldman Sachs Global Investment Research

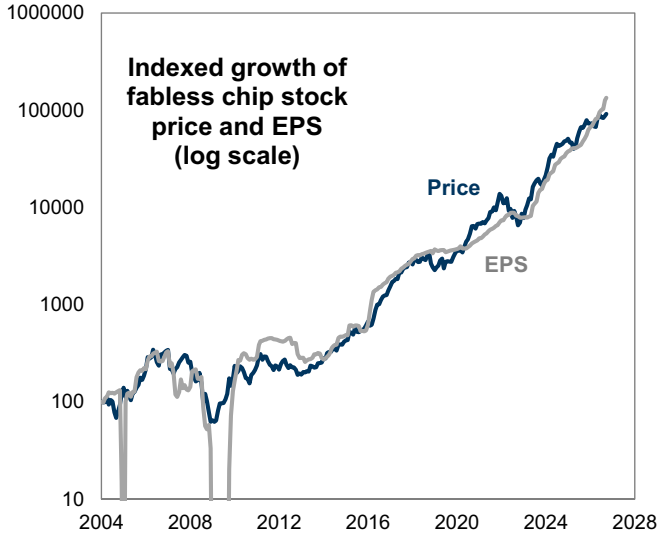
In contrast, the fabless chip stocks also trade at discounted valuations but have had less extreme margin expansion. Margins for the largest fabless chip stocks are only modestly above their historical average. Earnings in the next few years would need to decline by about 10% to bring the P/E back to its historical average. This is similar to the magnitude of historical realized cyclical earnings declines for the group.

Exhibit 16: P/E of largest fabless chip stocks
Fabless chip stocks represents dollar-weighted combination of NVDA, AVGO, AMD



Source: FactSet, Goldman Sachs Global Investment Research

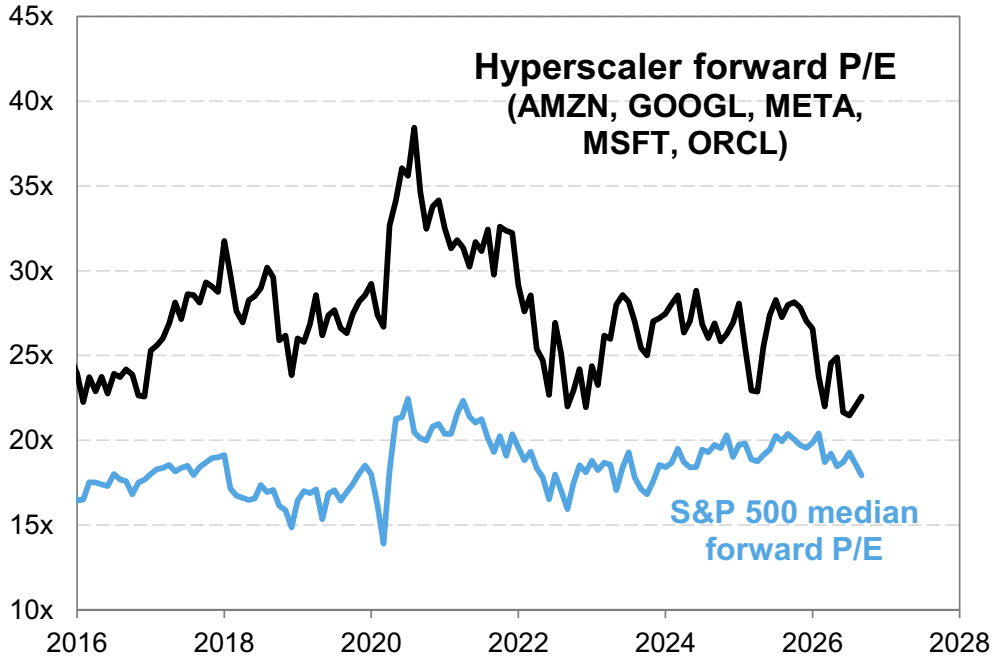
Exhibit 17: Price vs. EPS of fabless chip stocks
Fabless chip stocks represents dollar-weighted combination of NVDA, AVGO, AMD



Source: FactSet, Goldman Sachs Global Investment Research

The likelihood of decelerating capex growth should be a positive catalyst for the hyperscalers. We previously laid out three conditions for the hyperscalers to regain leadership. These include decelerating macroeconomic growth, demonstrated monetization of AI investment, and decelerating capex growth. Our economists believe US economic growth is no longer accelerating. The hyperscalers have already started to demonstrate signs of monetization, and we expect capex growth will slow in 2027. Uncertainty about the length, magnitude, and return on investment for this capex cycle will likely persist. However, the hyperscalers have de-rated to their lowest P/E multiples in over a decade, and the group now trades just four turns above the median S&P 500 company, its smallest premium on record. While investors have recently been focused more on the hyperscalers' negative free cash flows than on their earnings, visibility into improving FCF as revenues accelerate and capex decelerates should be supportive for the stocks.

Exhibit 18: Hyperscaler valuations have de-rated to lowest level in 10 years

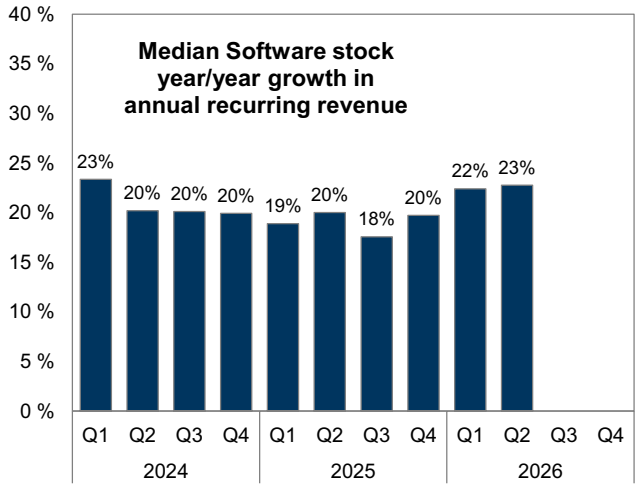


Source: Goldman Sachs Global Investment Research

We expect the progression of the AI trade towards application roll-out will remain a source of return dispersion. We believe the "winners" and "losers" from the application layer will take several quarters to play out, as AI applications get built and adoption broadens.

The threat of disruption had weighed on the share prices of many Software stocks earlier this year, but the group has rebounded sharply in recent weeks. Reported Software fundamentals have shown little sign of AI disruption, with annual recurring revenue growth for the median stock in GS analyst coverage accelerating slightly in recent quarters. A group of Software stocks has rallied by 45% and is now within 8% of its high reached in late 2025. Software stocks focused on IT infrastructure and cybersecurity have posted the strongest returns, while enterprise management software remain below their previous highs.

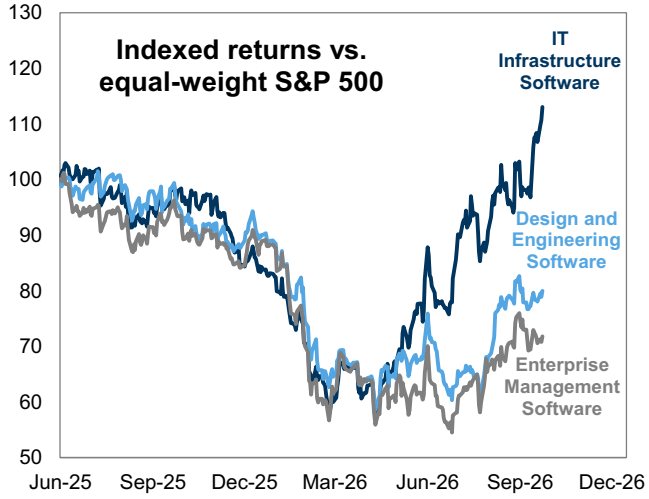
Exhibit 19: Software ARR has accelerated in recent quarters



Source: Goldman Sachs Global Investment Research

Exhibit 20: Within Software, IT infrastructure stocks have fared best

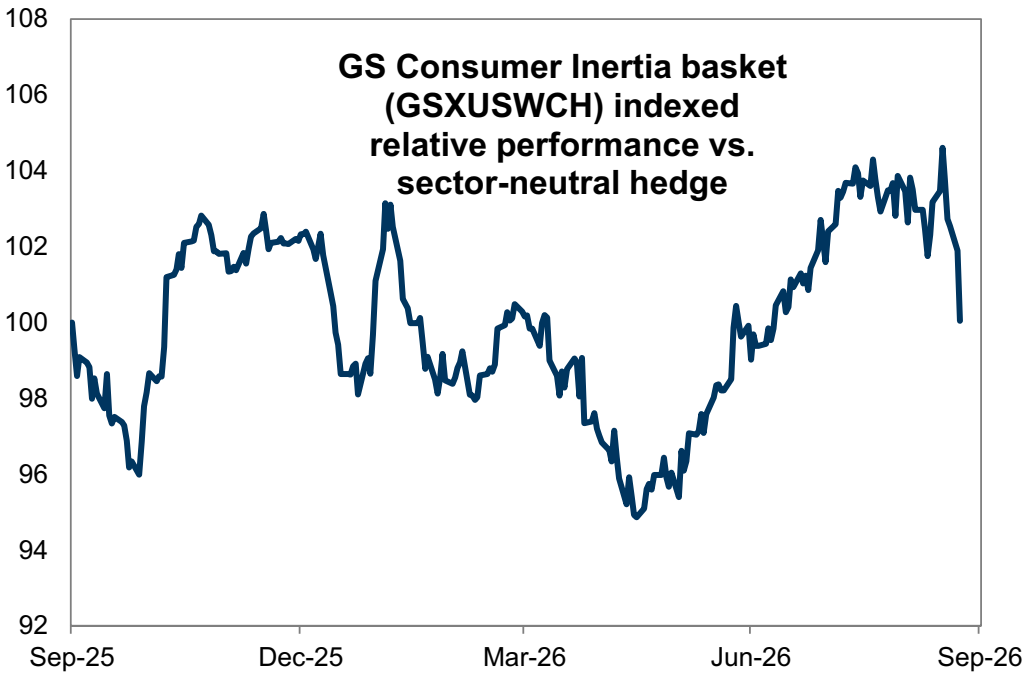
based on RBICS classifications



Source: FactSet, Goldman Sachs Global Investment Research

Consumer-facing businesses with high switching costs have also recently emerged as a potential area of disruption. The increasing use of agents, including by new consumer-facing tools such as Muse, may disrupt the business models of some consumer firms. Our colleagues in Global Banking & Markets developed a basket of "consumer inertia" stocks (GSXUSWCH) where user stickiness is driven more by effort required to take action rather than product differentiation. The basket includes insurance, telecom, travel, subscriptions, marketplaces, and fintech. The basket has lagged a sector-neutral hedge by 5 pp since mid-September.

Exhibit 21: Stocks with "consumer inertia" have underperformed recently

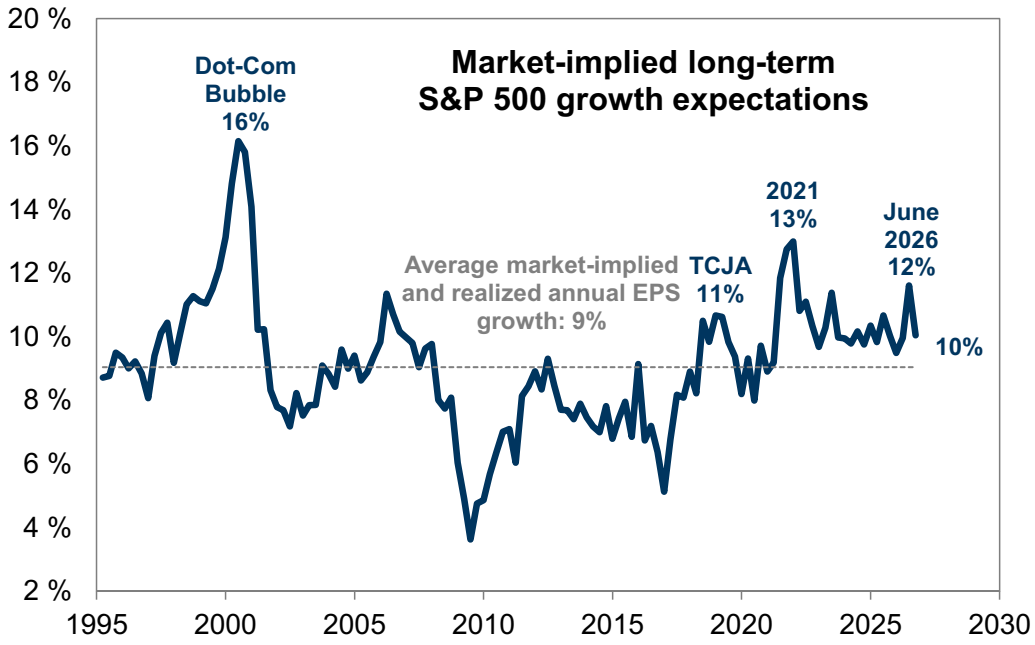


Source: Goldman Sachs Global Investment Research

Enterprise AI adoption remains in early stages but spending has recently accelerated, suggesting the impact of AI on corporate earnings should grow increasingly visible in coming quarters. The Census Bureau survey suggests that 40% of large US firms are using AI in the production of goods and services. Our analysis of Q2 earnings commentary suggests roughly 50% of S&P 500 companies are using AI in the context of productivity, but very few companies have quantified the productivity uplift today. However, the revenue run rates of model providers and alternative measures of company spending on AI have accelerated during the last several months alongside the increasing use of agentic harnesses.

In contrast with past periods of market exuberance, the equity market today does not appear to be pricing expectations of widespread productivity gains. Baskets of AI productivity beneficiaries have traded roughly sideways on a relative basis YTD, matching the trajectory of near-term forward EPS estimates. Similarly, the measure of market-implied long-term growth expectations in Exhibit 22 is only slightly above the historical average (for details, see US Equity Views: The next phases of the AI trade).

Exhibit 22: Market-implied long-term growth expectations are relatively contained



Source: Goldman Sachs Global Investment Research

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Reg AC

We, Ryan Hammond, Ben Snider, Daniel Chavez, Kartik Jayachandran and Christophe Sung, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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GLOBAL STRATEGY VIEWS

Competition for Capital

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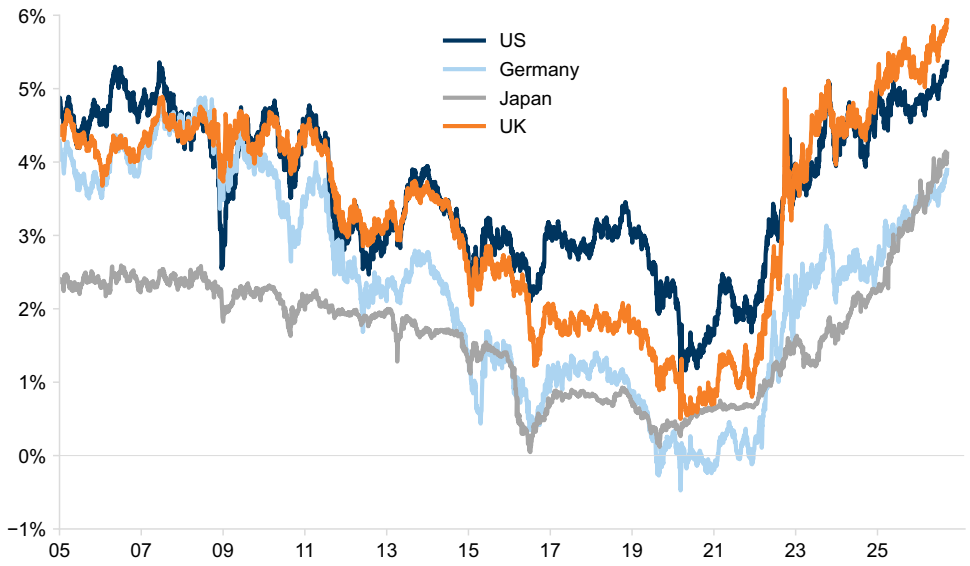
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1. There are two themes dominating our investor conversations: the impact of AI and the rise in interest rates. The two issues are linked as demand for capital from both the private and public sectors increase. In the private sector a surge in capex spending to fund AI infrastructure has eaten into free cash flow and forced companies to raise more in debt and equity markets. Meanwhile, government borrowing needs have increased as priorities shift towards upgrading critical infrastructure, energy security and defense at a time when cyclical inflationary pressures driven by higher energy prices are also resulting in higher policy rates. The combination has pushed up the cost of capital. As recently as 2022, for example, 30-year bond yields in Germany and Japan were close to zero (Exhibit 1). A rise in yields, together with more uncertainty (over geopolitics and the future impact of AI) have, in combination, pushed up the cost of capital.

Exhibit 1: Rising yields and uncertainty have increased the cost of capital
30-year government bond benchmark yields (%)

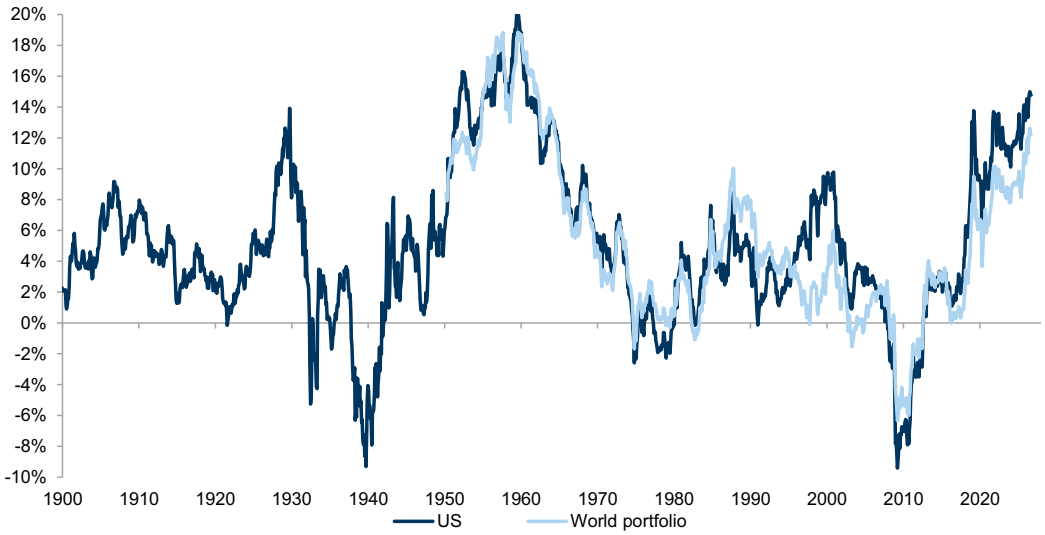


Source: Datastream, Goldman Sachs Global Investment Research

2. This rise in yields comes after a near-record period of equity outperformance relative to bonds over 10-year holding periods (Exhibit 2).

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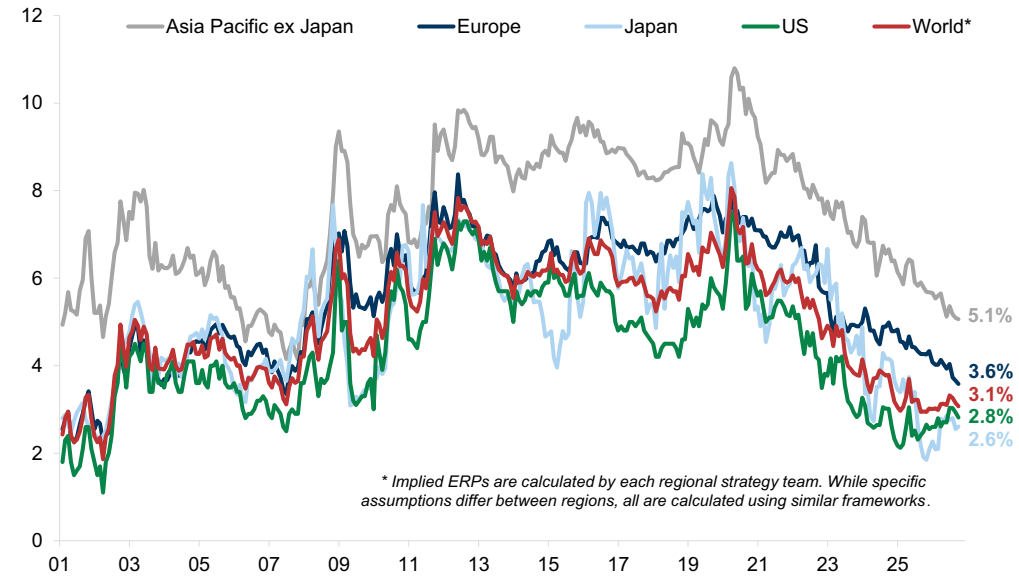
Exhibit 2: Equities have enjoyed a near record period of outperformance relative to bonds
10-year rolling annualised equity vs. bond performance (ex post equity risk premia)



Source: Robert Shiller, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

As equities have continued to outperform bonds, equity risk premia have fallen back to levels last seen in the late 1990s, leaving equity markets more vulnerable to further increases in bond yields. That said, in the US and Japan, the ERP has bounced off recent lows (Exhibit 3).

Exhibit 3: Equities have re-rated versus bonds amid higher inflation
Global market implied ERP (%)

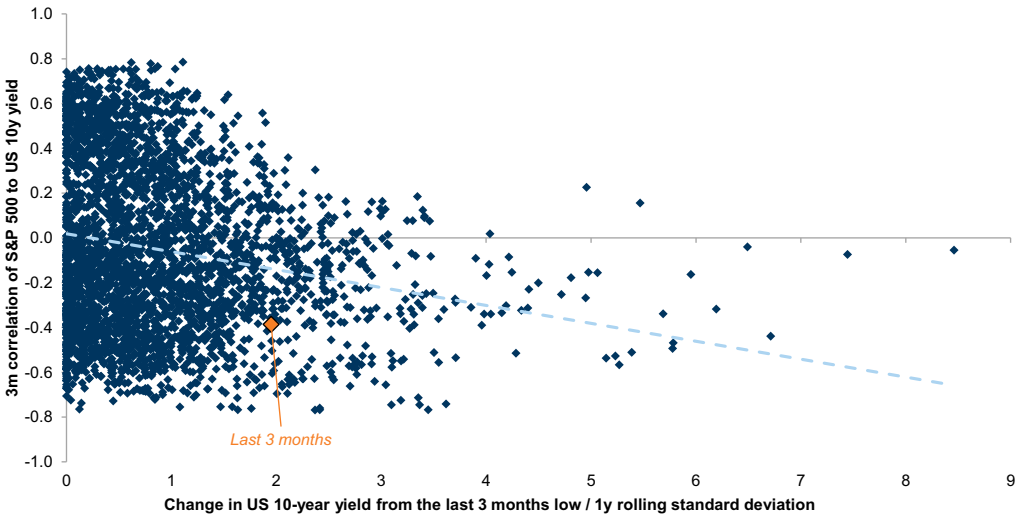


Source: Goldman Sachs Global Investment Research

3. The impact of rising bond yields, however, depends not only on the level of yields, but also on the reasons as well as the speed of the adjustment (Exhibit 4). It is the speed issue that has caused the biggest impact in the recent past. As our US strategists note, stocks have usually generated positive returns alongside rising interest rates unless the pace of rising rates exceeded 2+ standard deviations and this is a pattern that is broadly

reflected in other regions too. Today, a 2-standard deviation move in 10-year Treasury yields would equate to about 50bps over a month or 30bps over two weeks, similar to the adjustments in recent days which explains the pullback in equity prices.

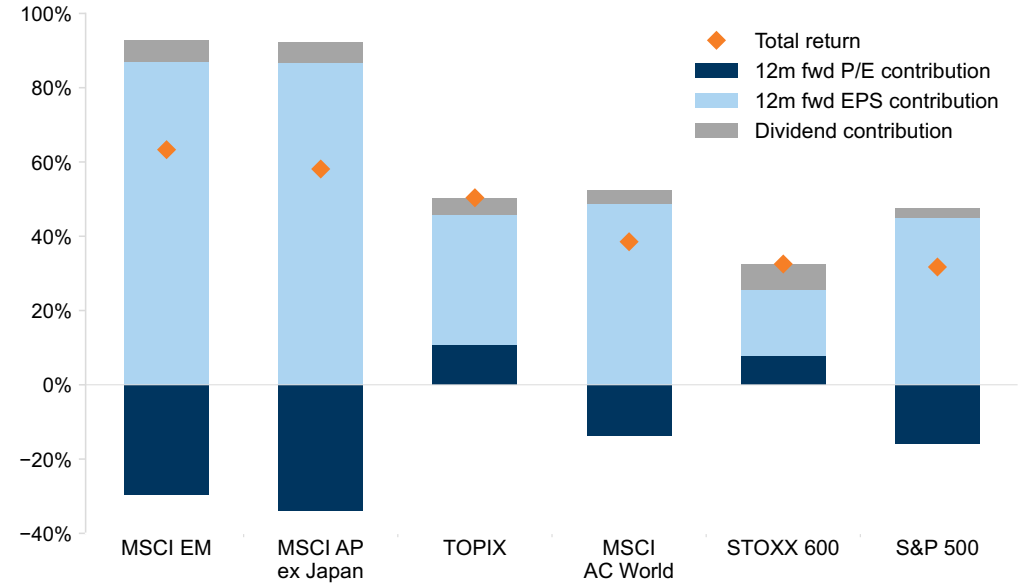
Exhibit 4: A continued sharp increase in bond yields could put pressure on equities
Data since 1962



Source: Datastream, Goldman Sachs Global Investment Research

4. But despite the rise in yields, nominal GDP and profit growth remain strong and, until recently, the impact of rising bond yields has been offset by the strength in corporate profits. Earnings growth has been the main driver of equity returns over the past 18 months in all regions (Exhibit 5). This strength has meant that PE multiples have either been flat (Japan and Europe) or have fallen (the US, Asia and EM).

Exhibit 5: Earnings have driven most of the returns since 2025
Total return contribution since Jan-2025 (local CCY)



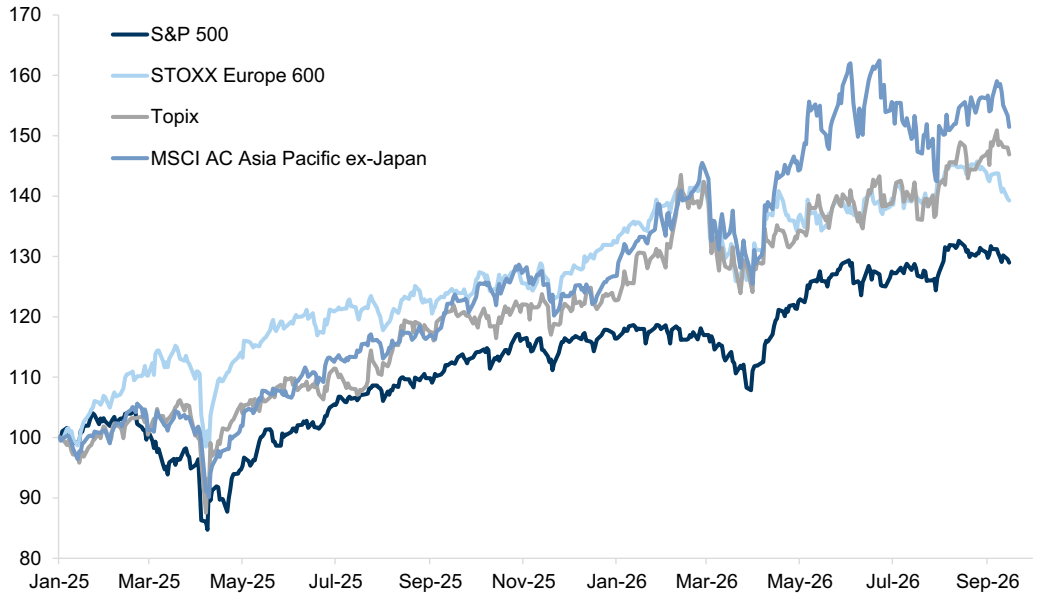
Source: Datastream, STOXX, Goldman Sachs Global Investment Research

In the US, in particular, the forward PE multiple for the S&P has come down from 22x at

the start of the year to 19x, in line with its long-run average, despite the market being close to its all-time high.

5. While US equities have driven the bulk of equity outperformance for much of the fifteen years post the financial crisis, the improvement in earnings across regions, and the sharper de-rating in the US, has resulted in equity performance being broader-based by geography. Since 2025, the US equity market has been the weakest of the major regions (Exhibit 6), a turnaround from the dominant trend since the financial crisis and increasing investor returns from a more geographically diversified portfolio.

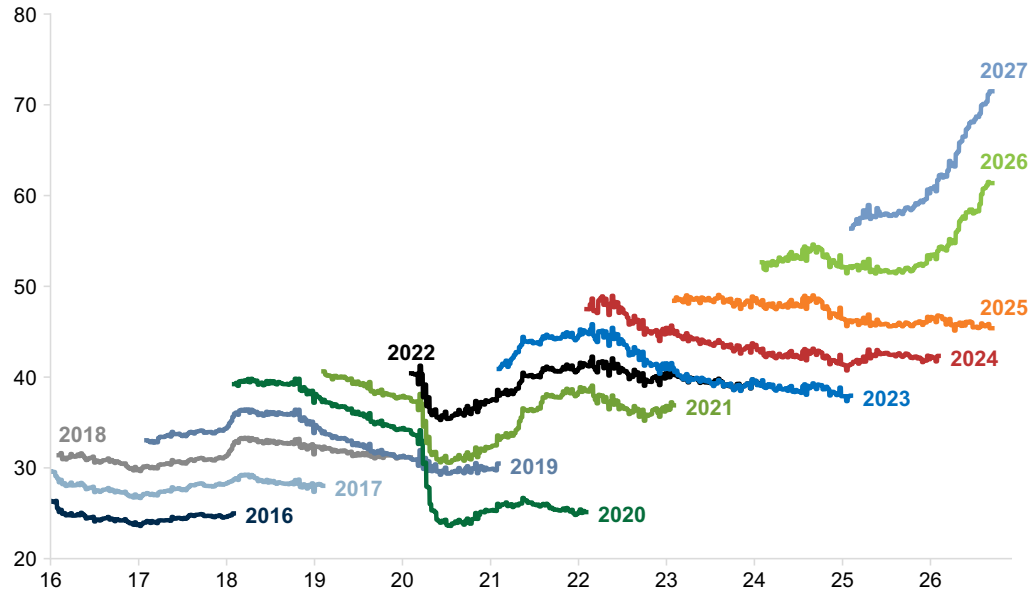
Exhibit 6: US equities have underperformed other major regions since 2025
Indexed price performance (USD)



Source: Datastream, STOXX, Goldman Sachs Global Investment Research

6. Supportive profit growth has been accompanied by positive earnings revisions, with 2026 and 2027 estimates moving higher across major regions. There are four broad areas which have been driving much of this profit growth. First, technology earnings continue to be very robust. Second, rising energy prices have pushed up profits in the commodity sector. Third, banks have generally enjoyed strong earnings backed by positive nominal GDP growth, steep yield curves and strong private sector balance sheets. Finally, sectors such as Industrials have benefited from the surge in AI capex spending which has spilled over into improved revenues for the 'pick and shovels' of AI infrastructure. The breadth of the earnings' growth has also increased the opportunity for investors to diversify across sectors as well as countries.

Exhibit 7: Earnings revisions have unusually been positive
Consensus estimates in USD. MSCI AC World EPS

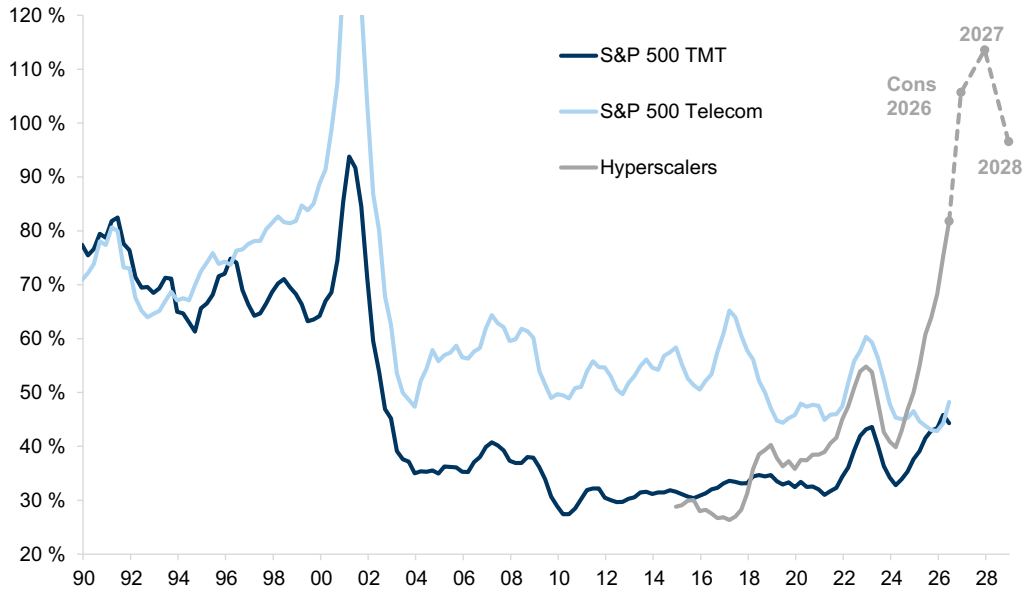


Source: Datastream, Goldman Sachs Global Investment Research

7. While technology profit growth has remained strong, the surge in capex spending among the hyperscalers has increasingly eaten through their free cash flow (Exhibit 8) prompting companies to look for alternative sources of funding, in the credit and equity markets. The capex spending growth for AA-rated issuers has been substantial: the 65% year-over-year growth in Q2 marks the 10th consecutive quarter that aggregate AA capex growth exceeds 35%. They have also turned to the convertible bond market where volume has also increased year to date reaching $135 billion in the US, with AI-related borrowers driving 44% of total issuance, on our estimates.

Our Credit team raised their full-year US IG gross issuance forecast by $200bn to $2.3 trillion and AI-related issuers account for a quarter of USD IG gross supply this year, pushing overall supply to a year-to-date record.

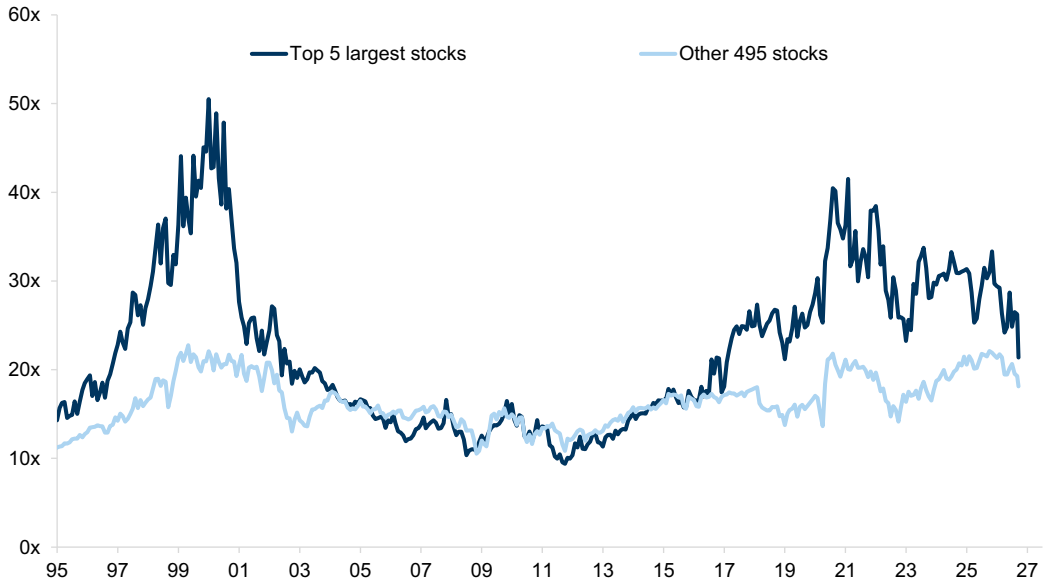
Exhibit 8: CAPEX is concentrated in few companies
Hyperscaler capex as % of cash flow from operations, dashed line representing 2026 consensus



Source: Compustat, FactSet, Goldman Sachs Global Investment Research

The combination of higher cost of capital and greater capital intensity in the tech sector has reduced the value of their future cash flows and triggered a de-rating in the biggest companies. The dominant capex hyperscalers in the US now have a forward PE close to the average of the rest of the market.

Exhibit 9: Large-cap valuations are less demanding than perceived
S&P 500 12m forward P/E multiple

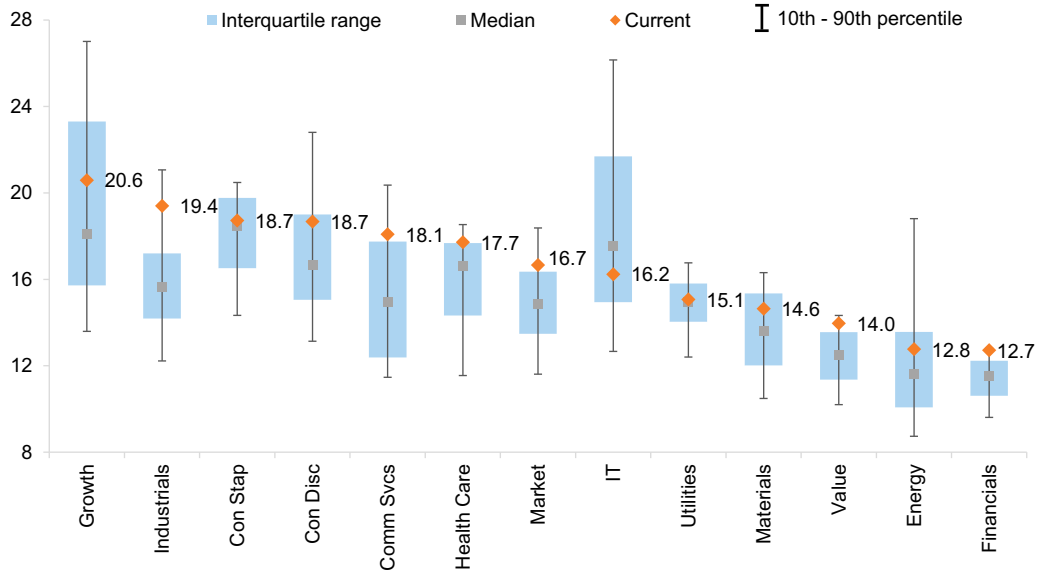


Source: Datastream, Goldman Sachs Global Investment Research

8. Software companies have also de-rated on the back of fears of disruption from AI while memory and chips stocks have also experienced lower multiples as their earnings

surge. Globally technology has seen its PE ratio moderate, falling below its 20-year median (Exhibit 10). Meanwhile, the spending of these companies has spilled over into better earnings and more visible cash flows in many other parts of the economy. Healthcare, Consumer Discretionary and Industrials are all more expensive in absolute terms than IT, and Industrials has re-rated and now has a PE close to the 90th percentile of its 20-year range.

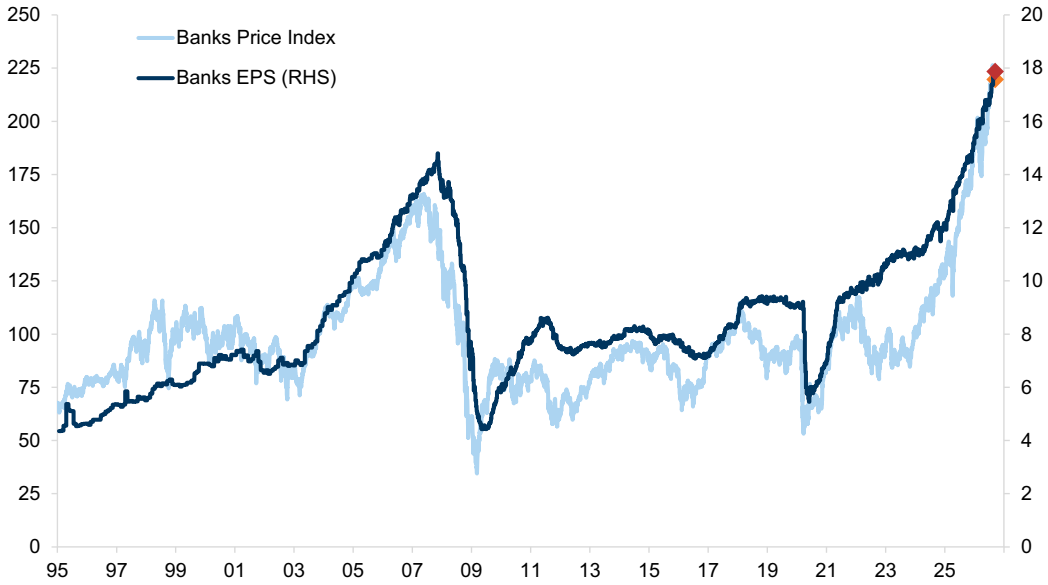
Exhibit 10: Industrials now trade at higher valuations than Technology
MSCI World sector/style valuations, 12-month forward P/Es relative to the last 20 years



Source: FactSet, Datastream, Goldman Sachs Global Investment Research

9. For these reasons, we have argued that Technology equities have not experienced a classic bubble because valuation levels are much lower than in previous financial bubbles. Nevertheless, the risk for investors is less about valuation and more whether there is an unsustainable earnings bubble. Bubbles in earnings, rather than valuations, have occurred in previous periods. The Bank sector, for example, briefly became the biggest sector in the S&P 500 in the run up to the financial crisis of 2008/09 (Exhibit 11). Unlike the technology bubble of the late 1990s, or the Japanese bubble of the late 1980s, Bank stocks did not experience a major valuation bubble at the time, but the surge in earnings that drove the sector's outperformance proved to be unsustainable.

Exhibit 11: Banks peaked again on earnings momentum
MSCI World Banks



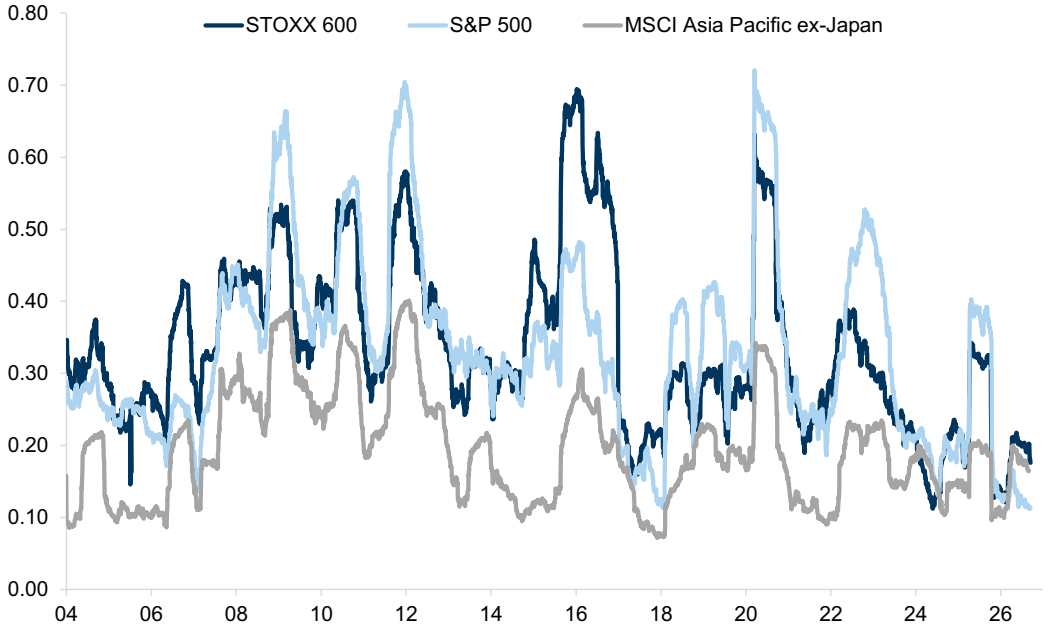
Source: Datastream, Goldman Sachs Global Investment Research

There are, however, important differences to the Tech sector today. The earnings of the banks were boosted by sharply rising leverage and the asset that they were financing – US real estate – did experience a valuation bubble. When the housing market collapsed, pushing economies into recession, it triggered a sharp fall in underlying earnings for the banking sector. There are important differences today. First, technology profits remain very robust, and balance sheets are strong overall. Second, interest coverage ratios for the aggregate S&P 500 and its median stock, for example, rank in the 99th and 68th percentiles relative to the past 20 years. Third, demand for compute is accelerating and outstripping supply; Microsoft recently declared their intent to triple data centre capacity in 6 years. The bottlenecks are increasingly power, land and data centre capacity rather than a lack of compute demand. At our annual Communacopia Technology Conference, Nvidia reiterated its 2030 AI TAM outlook to $3tn-$4tn and supply constraints continue to be broad based. Nonetheless, any slowdown in profit growth, in an environment of a much higher cost of capital, could put downward pressure on equity prices, reducing confidence in future cash flows across the ecosystem from the hyperscalers to the 'pick and shovels' that have been benefiting from the capex boom.

10. Given the near-term risks in bond markets, we remain neutral on equities, alongside other asset classes on a 3-month horizon, but continue to be overweight equities over 12 months. While profit growth is likely to slow from the recent strong pace, ongoing growth in nominal GDP should underpin equity markets and the earnings cycle propelling equity returns even without any valuation expansion from current levels and our economists continue to see a low risk of recession. As our asset allocation team has shown, earnings growth and ROE are likely to continue to be the key drivers of equity performance relative to bonds, and we expect these fundamentals to remain supportive for equity returns. Nevertheless, we continue to recommend well diversified equity exposure as opportunities continue to broaden across geographies, sectors and factors.

Stock correlations are at multi-year lows across the major equity markets (Exhibit 12).

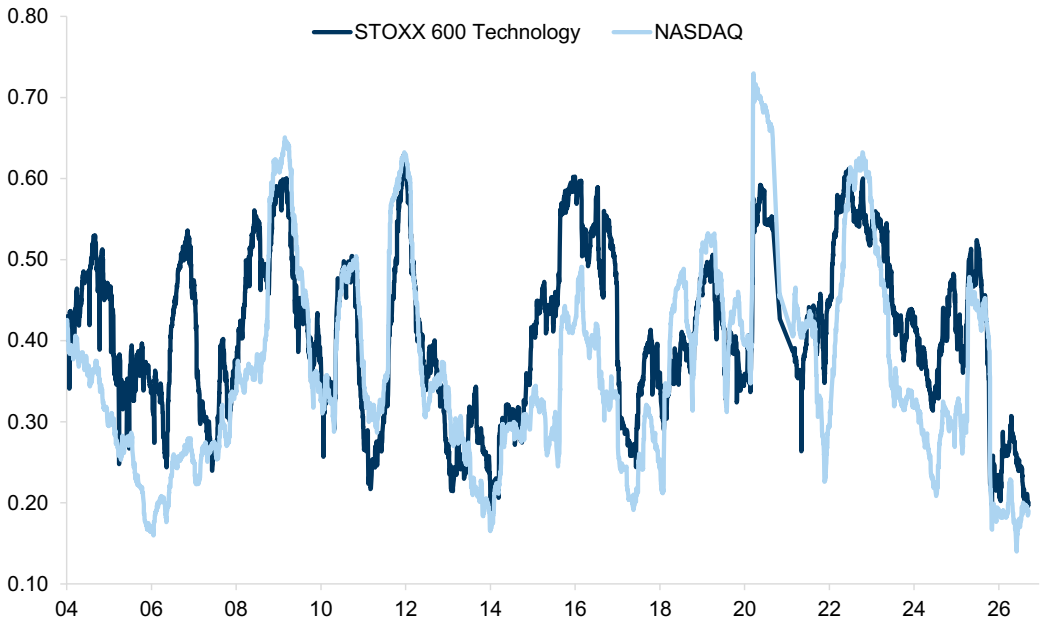
Exhibit 12: Pairwise correlation is low across regions
6-month average pairwise correlation



Source: Bloomberg, Goldman Sachs Global Investment Research

Even within sectors like technology, correlations have fallen sharply (Exhibit 13). The 'magnificent seven' that was long seen as a homogeneous group of technology winners has also seen significant dispersion of returns. Differentiation, diversification and alpha generation are likely to remain the key drivers of equity returns. With higher bond yields, valuations are not likely to rise making earnings the critical driver of returns for investors.

Exhibit 13: Correlations have fallen sharply within Technology
6-month average pairwise correlation



Source: Bloomberg, Goldman Sachs Global Investment Research

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1 September 2026 | 4:10PM EDT

AI ADOPTION TRACKER

August 2026: Adoption Rises to 22.4%

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- We update our AI adoption tracker through August 2026.
- **AI-related investment growth** remains strong, particularly for semiconductors, where equity analysts expect global revenue to reach $863bn (ann.) by the end of 2026. 2026 gross profit forecast revisions since 2022Q3 for AI-exposed global public companies currently stands at $1.1tn. AI-related hardware investment in the US national accounts also continues to rise and now stands at $485bn (3-month annualized average; 1.5% of GDP) above its 2022 level. AI-related hardware shipments from Taiwan have grown by 76% since 2022 and US AI hardware net imports ticked up to $55.9bn in June.
- **AI adoption by firms** rose by 0.9pp to 22.4% among US establishments (with adoption expected to rise to 25.9% in the next 6 months) according to the Census Bureau's Business Trends and Outlook Survey (BTOS). Information is leading sector adoption at nearly 50%. At a detailed subsector level, computing and publishing firms continue to lead adoption, and warehousing and miscellaneous manufacturing firms report the largest expected increase in adoption over the next six months. Establishments with over 250 employees continue to lead in adoption (40%; latest six survey average) while adoption has grown more slowly recently among smaller firms. Other business surveys note that 40% of surveyed organizations have begun deploying and scaling AI agents.
- **AI's labor market impact** remains visible but narrow. We continue to observe employment drags in specific industries where AI-use cases have been established such as marketing, graphic design, customer service, and some tech occupations. However, this is offset by construction job growth in data center-exposed categories, which is up 209k since 2022 (relative to broader construction employment trends). The tech share of total employment continues to decline relative to the pre-2022 trend. The correlation between AI adoption and broad labor market slack measures remains limited but has strengthened in recent months. Corporate layoffs attributed to AI affected 11k employees in June, bringing the 2026 YTD total to 113k. 22% of Russell 3000 companies mentioned AI and labor-related keywords in 2026Q2.
- We continue to observe large **impacts on labor productivity** in the limited areas where generative AI has been deployed but only modest signs of economy-wide impacts. Academic studies imply a 23% average uplift to productivity, while company anecdotes imply slightly larger efficiency gains of around 32%.

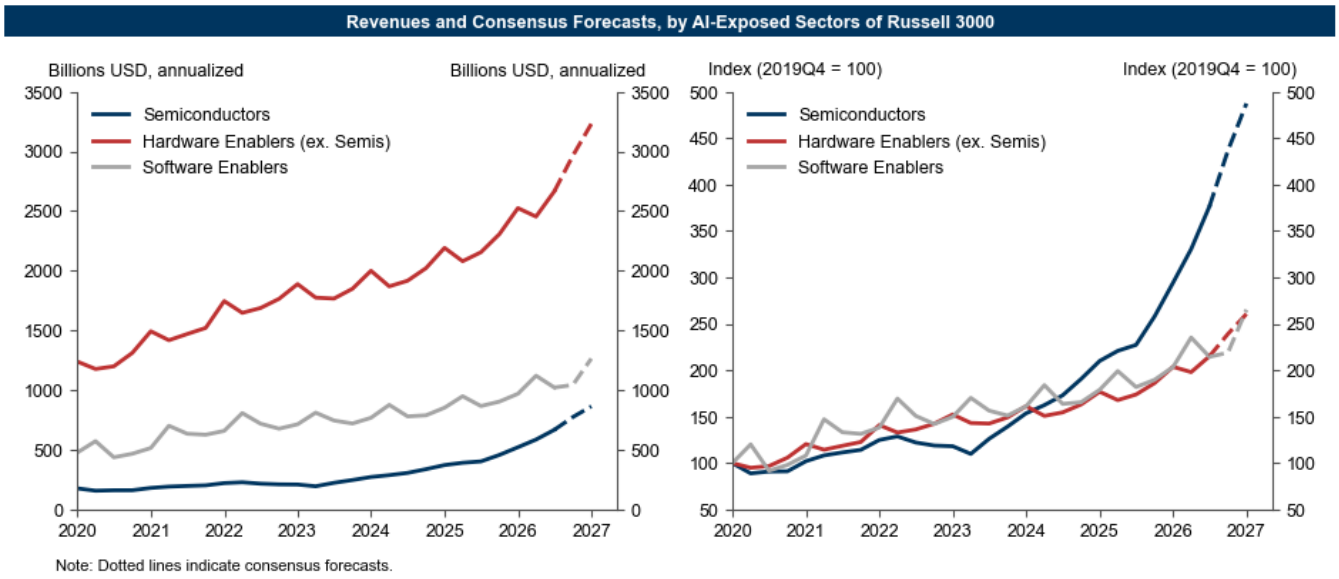
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Industries with higher AI adoption rates are now showing a slight acceleration in productivity growth over the past year in official US data.

August 2026: Adoption Rises to 22.4%

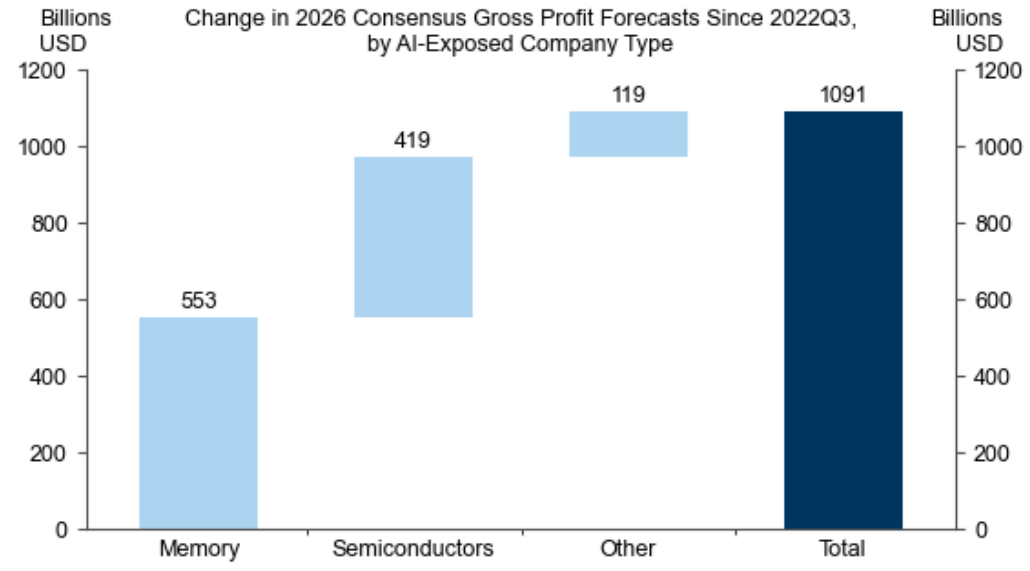
AI Investment

Exhibit 1: Semiconductor Company Revenues Are Projected to Rise to $863bn (Annualized; vs. 2022) by End-2026



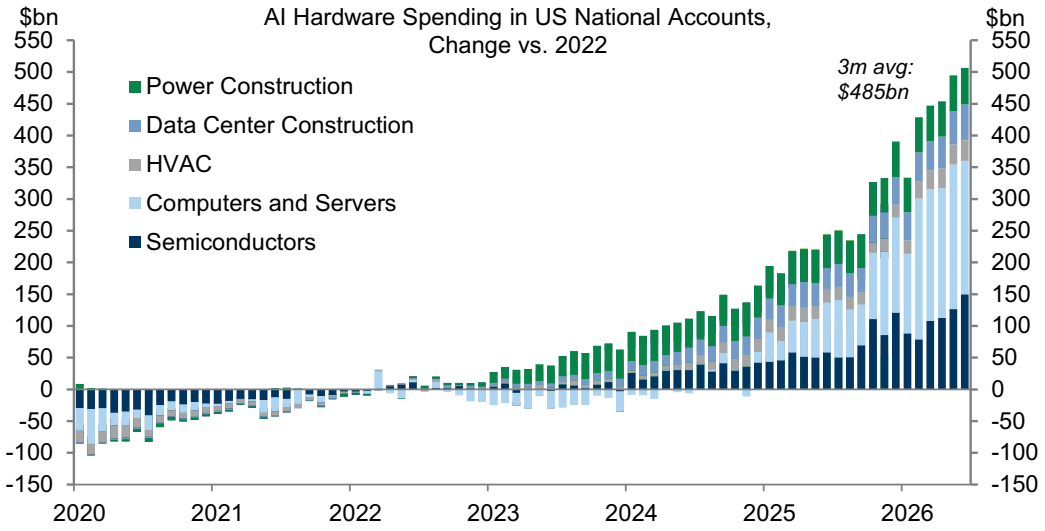
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 2: AI-Exposed Gross Profit Forecast Revisions For 2026 Currently Stand at $1.1tn



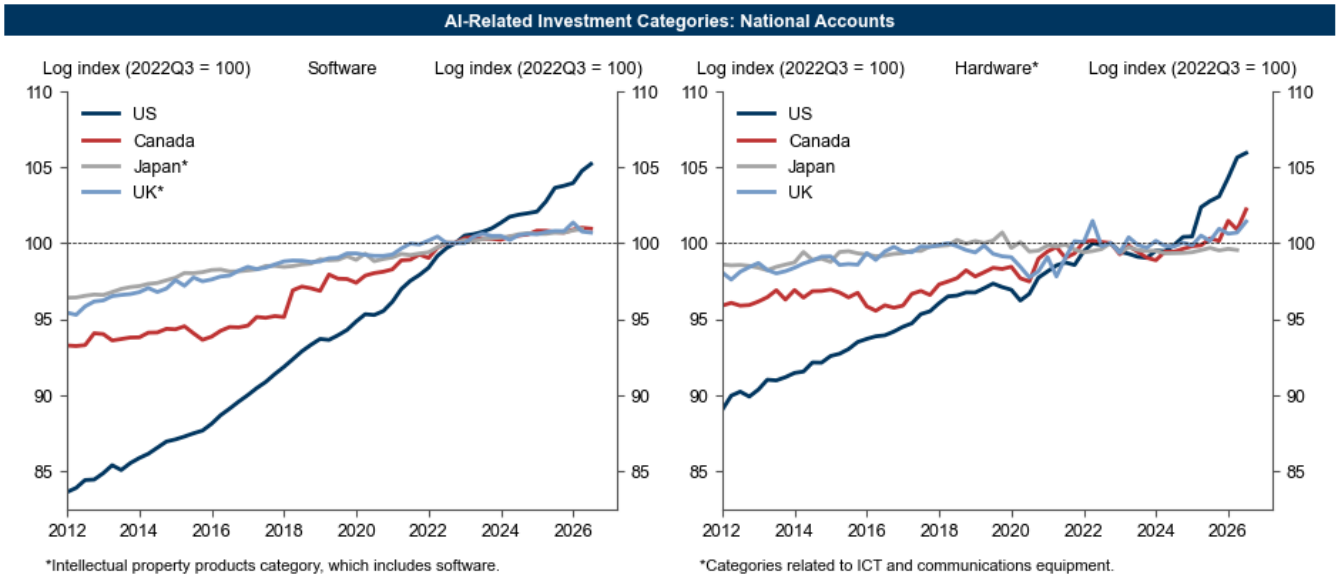
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 3: AI Hardware Investment in the US National Accounts Has Increased by $485bn Since 2022 on a Three-Month Annualized Basis



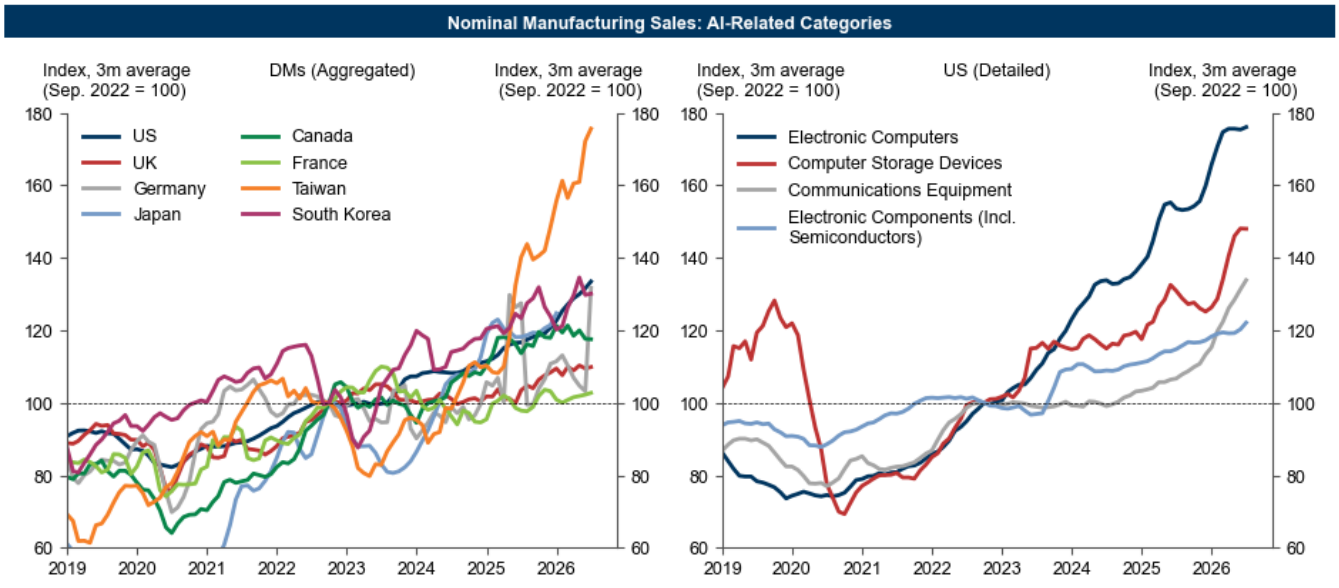
Source: Census Bureau, Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 4: US Investment in AI-Related Hardware and Software Ticked Up in Q2



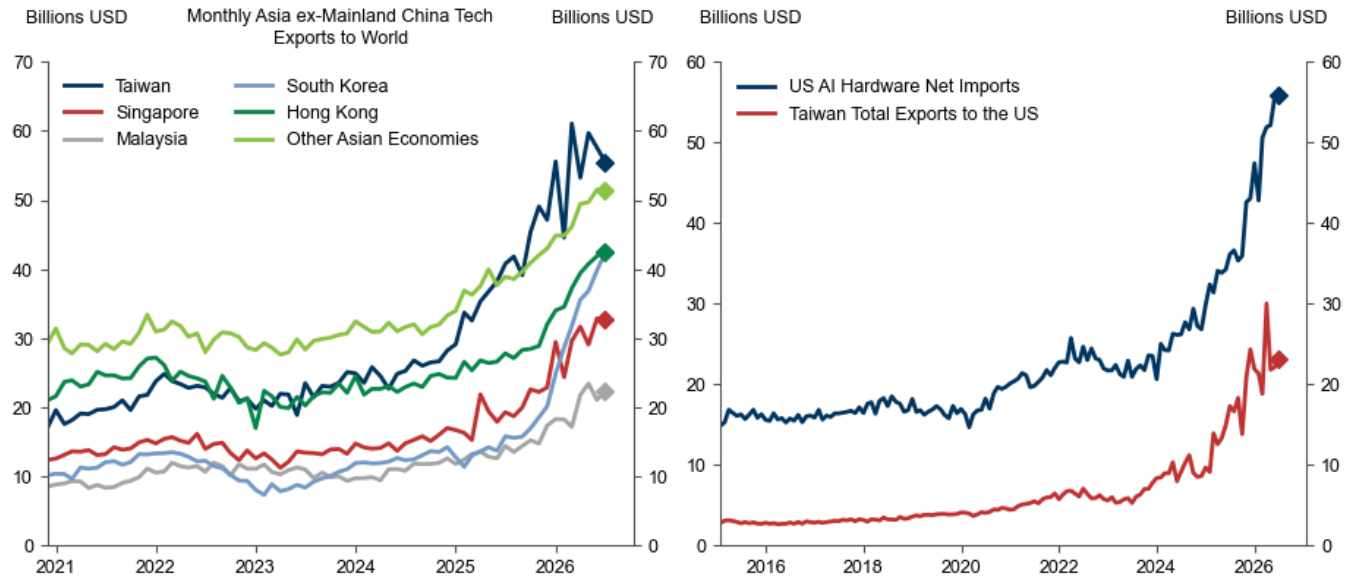
Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 5: AI-Related Manufacturing Shipments From Taiwan Have Grown by 76% Since 2022



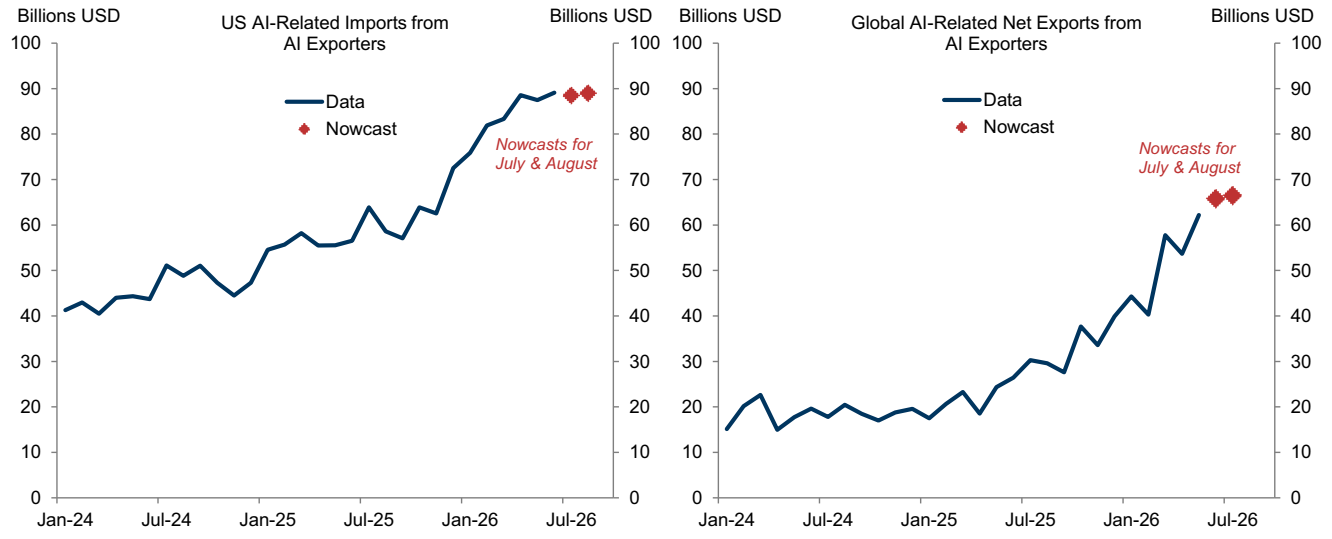
Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 6: US AI Hardware Net Imports Stood at $55.9bn in June



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 7: We Expect $66bn in Global AI-Related Net Exports from AI Exporters in August

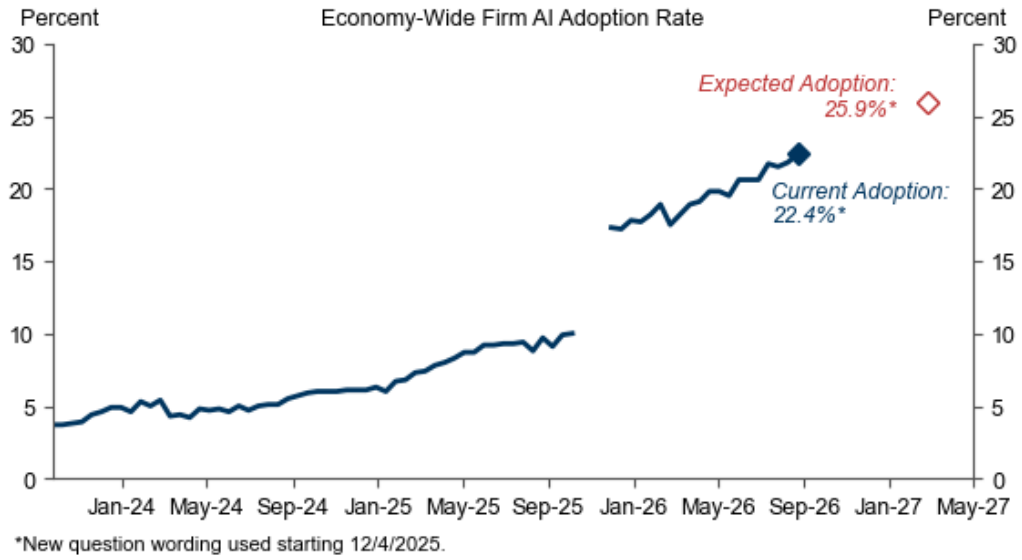


Source: Goldman Sachs Global Investment Research

AI Adoption by Firms

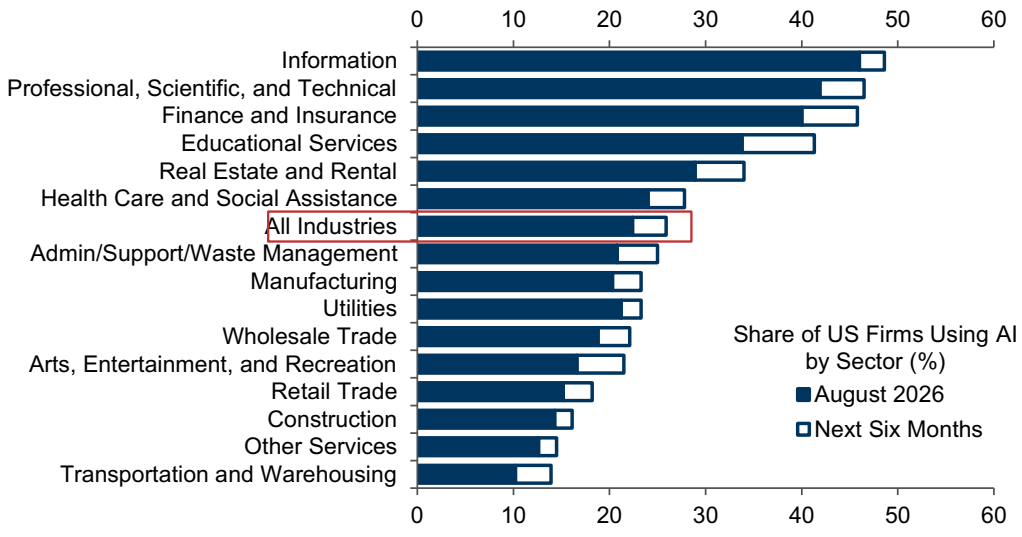
Note: The Census Bureau has rephrased questions from asking about AI use in "producing goods and services" to asking about AI use in "any business functions" starting from the December 4, 2025 release. There was a level shift observed in conjunction with the new question wording, and we will be tracking adoption rates under the new methodology going forward.

Exhibit 8: 22.4% of Firms Are Using Artificial Intelligence in Their Regular Business Functions (+0.9pp From July); 25.9% Expect to Use AI In the Next 6 Months



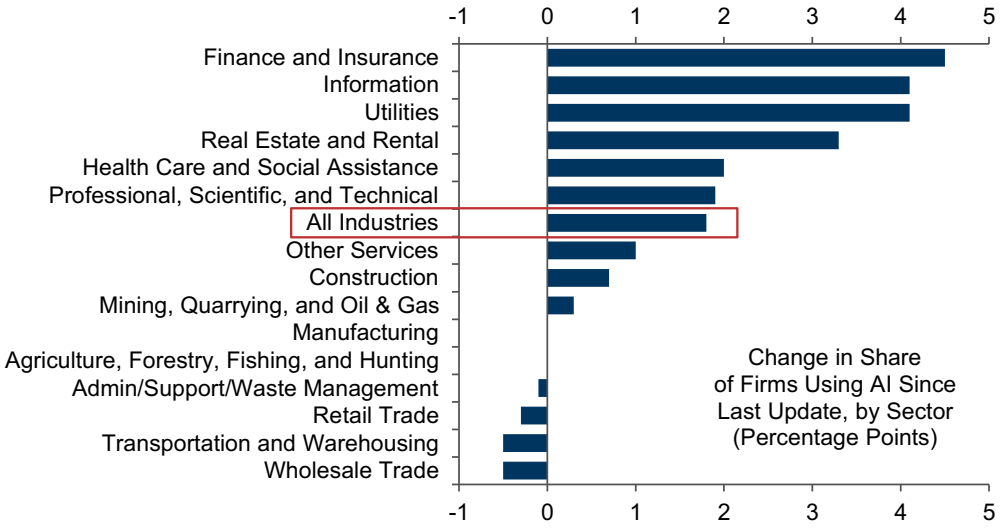
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 9: Adoption Is Approaching 50% in the Information Sector



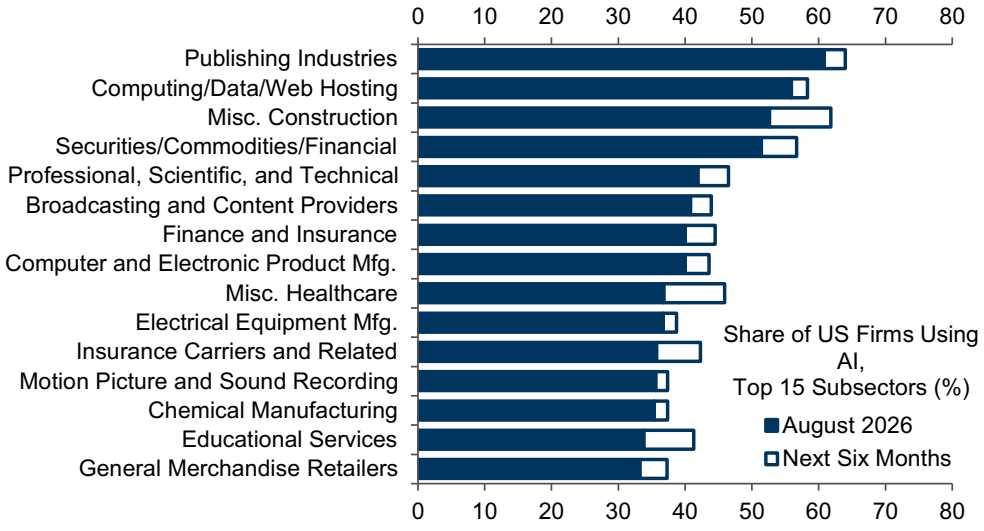
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 10: Finance and Insurance Firms Had the Largest Gains in Adoption Since Our Last Update



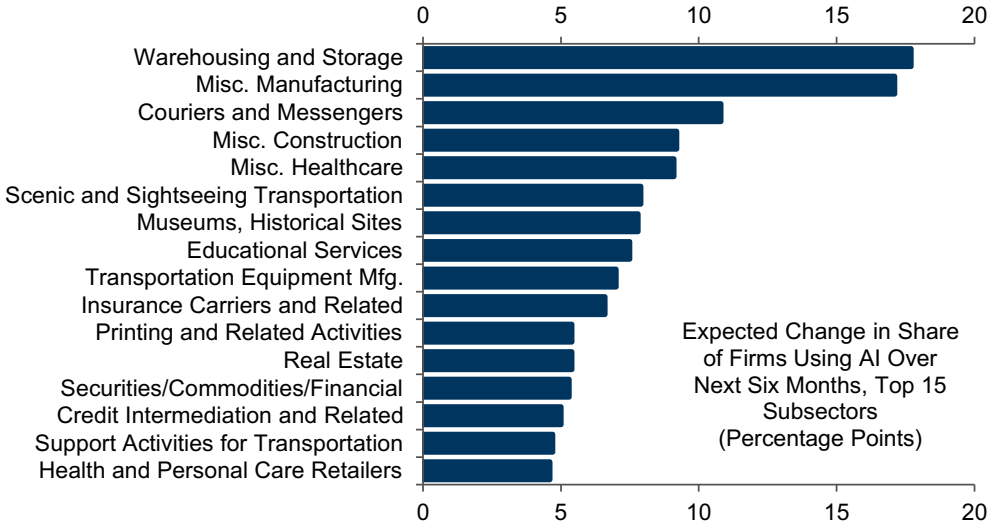
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 11: Publishing and Computing Continue to Lead Adoption at the Subsector Level



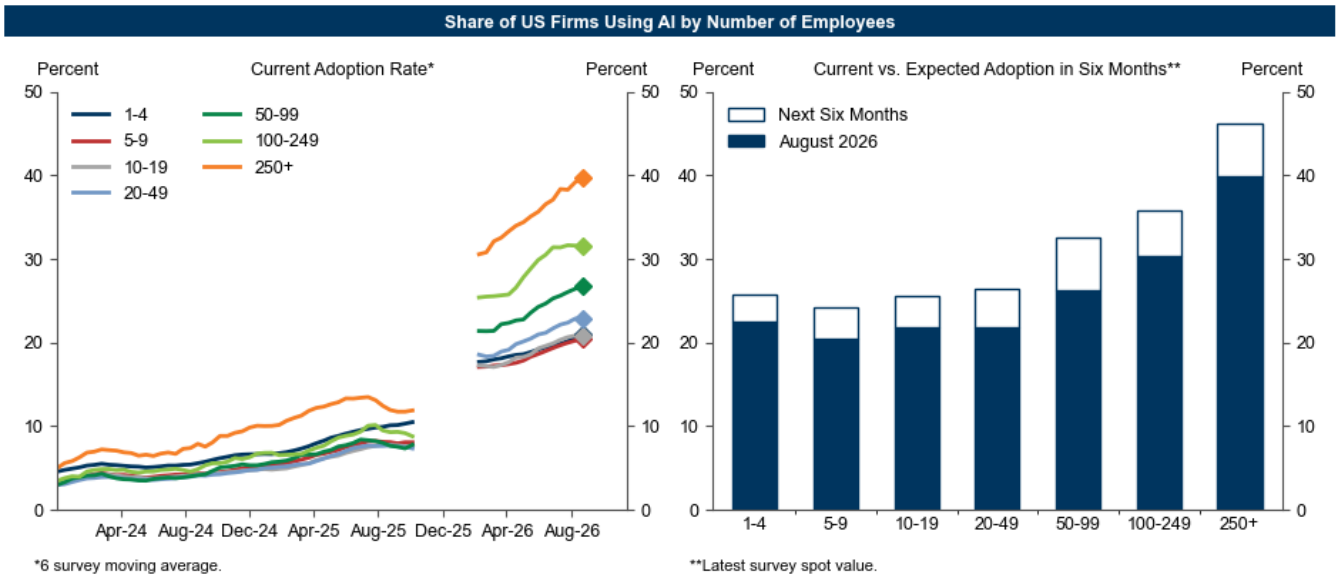
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 12: Warehousing and Storage and Miscellaneous Manufacturing Firms Report the Largest Expected Adoption Increases in the Next 6 Months



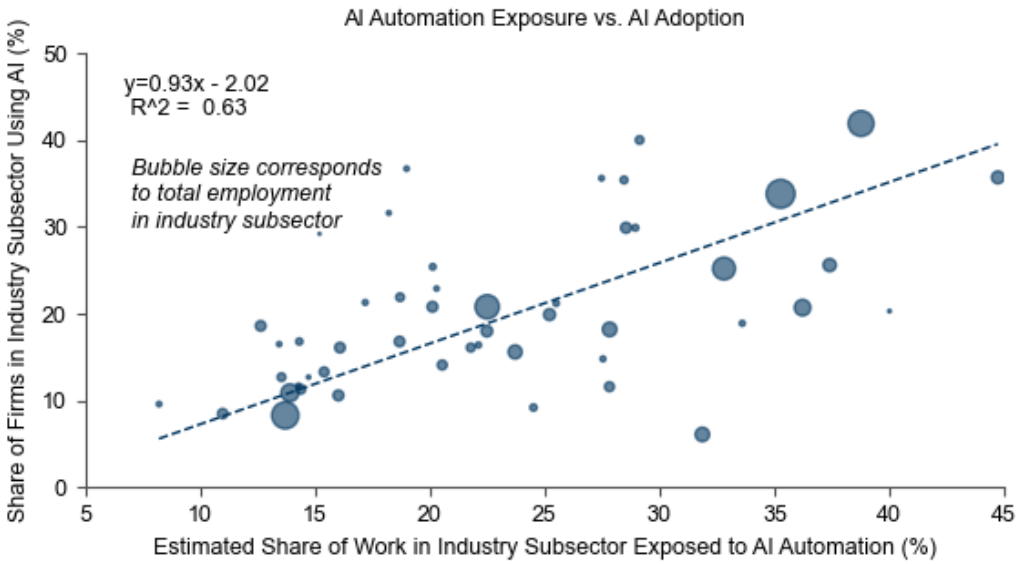
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 13: 40% of Large Firms With 250+ Employees Expect to Be Using AI in Six Months



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 14: AI Adoption Remains Strongly Correlated With Our Subsector AI Automation Exposure Scores



Share of work exposed to AI automation is a GS estimate based on the methodology in "Global Economics Analyst: The Potentially Large Effects of Artificial Intelligence on Economic Growth" (March 26, 2023).

Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 15: 40% of Surveyed Organizations Have Begun Deploying and Scaling AI Agents

Source	Date	Current Adoption / Impacts	Future Adoption / Impacts
McKinsey	August 2026	40% of large organization respondents have adopted AI agents and are scaling their deployment throughout the organization, up from 27% last year. Nearly a third of respondents reported that they had used agentic coding tools to build software in-house instead of purchasing it. 80% of respondents said AI has improved their individual productivity.	60% of surveyed organizations expect to increase their AI investments in the year ahead. 39% of respondents expect AI-related declines in their organization's total employment in the coming year.
Gallup	July 2026	47% of surveyed US employees in Q2 said their organization has integrated AI tools to improve workplace productivity, up from 41% last quarter. Individual AI use at work also increased to 52% of respondents. Among AI users, the most common uses of AI are writing and editing (51%), search or research (49%), and general assistance or problem-solving (39%).	The highest productivity ratings come from employees using AI for coding assistance and automation or process automation. Employees who use AI for a greater variety of tasks were also more likely to report that AI positively affects their personal productivity. Although corporate adoption and AI awareness has grown notably, one in five employees remain unsure if AI has been integrated at their organization
WRITER	April 2026	52% of employees say they've used AI agents over the past year. Employees reported saving around 6 hours per week with AI tools, while executives reported saving twice as much time at nearly 12 hours per week. The top use cases include editing and proofreading, personalized communication, and data analysis and insights. Few surveyed leaders reported seeing significant ROI from generative AI (29%) or AI agents (23%).	69% of executives report doing layoffs due to AI, but 39% admit their company doesn't have a formal strategy to drive revenue from AI.
Harvard Business Review and Appian	April 2026	Most organizations have moved beyond the experimentation phase with AI and 58% of surveyed organizations have AI in production. 69% of respondents reported that legacy systems are limiting their organization's ability to scale AI. Most respondents are seeing some returns from AI investments, but only 16% report realizing a high degree of measurable value.	Many organizations are currently exploring agentic AI systems (under consideration by 62% of respondents and currently used by 25%). Although nearly all organizations agree that clear guardrails are needed for AI agent adoption, fewer than half say their organization has defined such rules.
Gallup	April 2026	Half of all surveyed adults reported using AI in their jobs and 27% of employees in AI-adopting organizations say that their workplace has changed in disruptive ways due to AI in the past year. 65% of adopters said that AI has improved their productivity and efficiency, regardless of how often they personally use AI at work.	Eighteen percent of all U.S. employees say it is very or somewhat likely their job will be eliminated within the next five years due to AI or automation. Among employees working in organizations that have adopted AI, that share rises to 23%. However, employees in AI-adopting organizations were more likely to report both current expansions and reductions in headcount.
NVIDIA	March 2026	64% of respondents said their organizations are actively using AI in their operations. 28% said they're still in the assessment phase, while 8% said they're not using AI and have no plans to. Regional adoption among respondents was 70% in North America, 65% in EMEA, and 63% in APAC. Larger companies (with over 1000 employees) demonstrated broader adoption, deployed more use cases, and reported greater ROI. More than half of respondents (53%) said improved employee productivity was one of the biggest impacts AI had on business operations.	86% of respondents said their AI budgets would increase this year and nearly 40% said budgets would increase by 10% or more. Respondents reported that workflow and operations and retaining the right expertise to scale AI solutions were the key challenges to adoption.
Deloitte	January 2026	In the latest survey, 34% of companies were starting to use AI to deeply transform their businesses, 30% were designing key processes around AI, and the remaining 37% were using AI at the surface level with little or no change to underlying business processes. 84% of companies have not redesignated jobs or work tasks around AI capabilities. 25% of surveyed leaders reported that AI is having a transformative effect on their organizations, more than double from 12% a year ago.	Survey results showed that insufficient employee skills were the largest barrier to integrating AI into existing workflows. Nearly three-quarters of surveyed companies expect to be using agentic AI at least moderately in two years.
Gallup	December 2025	The percentage of US employees who reported using AI at work at least a few times a week increased from 19% in 2025Q2 to 23% in Q3. Unsurprisingly, employees working in knowledge-based jobs (such as tech or professional services) were more likely to use AI than those in frontline positions. Most common use cases include using AI to consolidate information or data, idea generation, and automating basic tasks. Users most often use AI chatbots and virtual assistants.	Gallup research has shown that employee AI adoption is strongly associated with having greater managerial support for AI and strategic integration of AI in their role, indicating that a strategic people-first approach is needed for effective AI adoption.

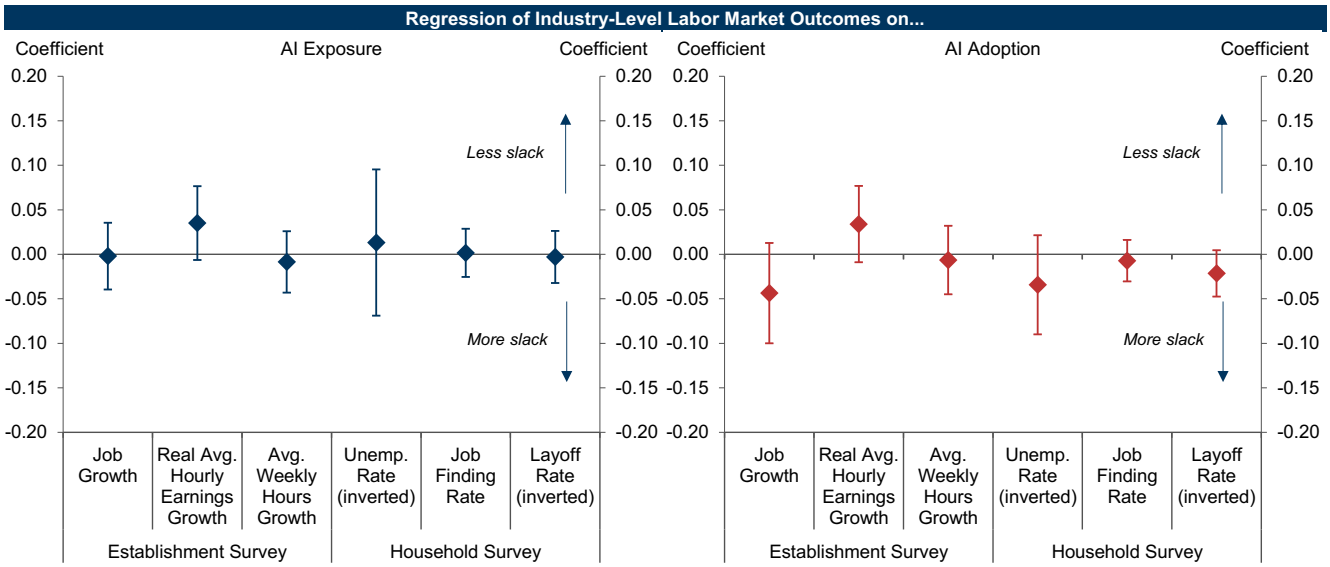
Green shading indicates surveys added since the last update.

Source: Goldman Sachs Global Investment Research

AI's Impact on the Labor Market

Labor Outcomes

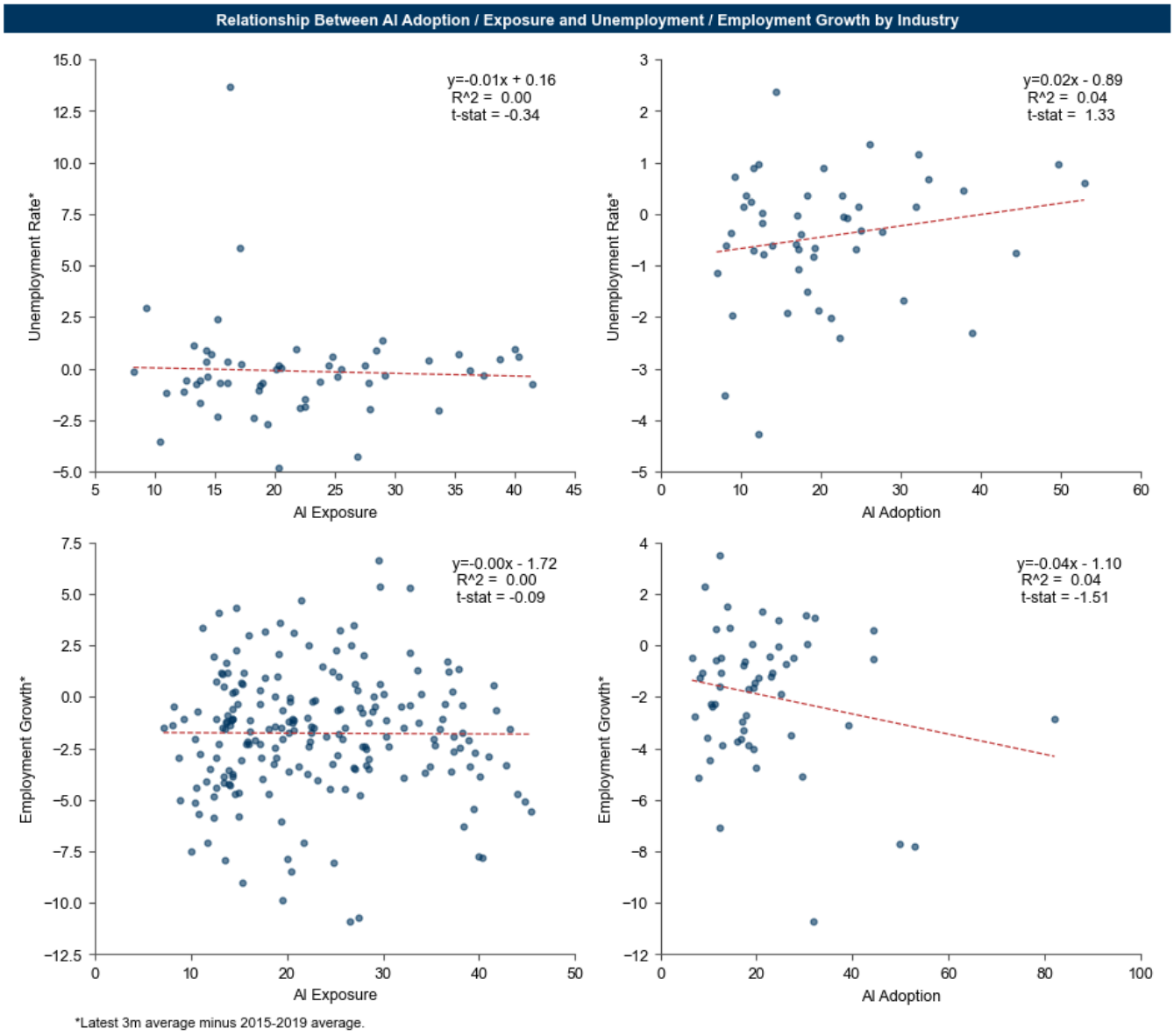
Exhibit 16: Some Signs of a Negative Relationship Between AI Adoption and Labor Market Slack Indicators, Particularly Job Growth



Note: All dependent variables are taken as the deviation from 2015-19 averages. Average coefficient over the past three months is plotted.

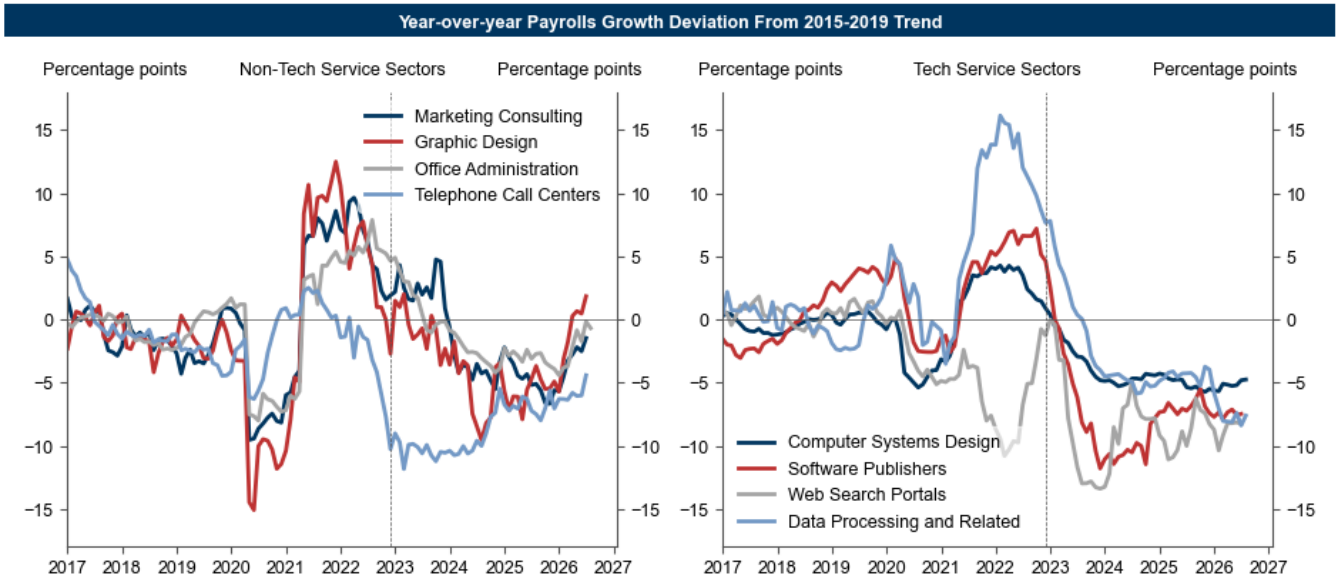
Source: IPUMS, US Bureau of Labor Statistics, Census Bureau, Goldman Sachs Global Investment Research

Exhibit 17: Recent Negative Relationship Between Adoption and Employment Growth, but Still Not Statistically Significant



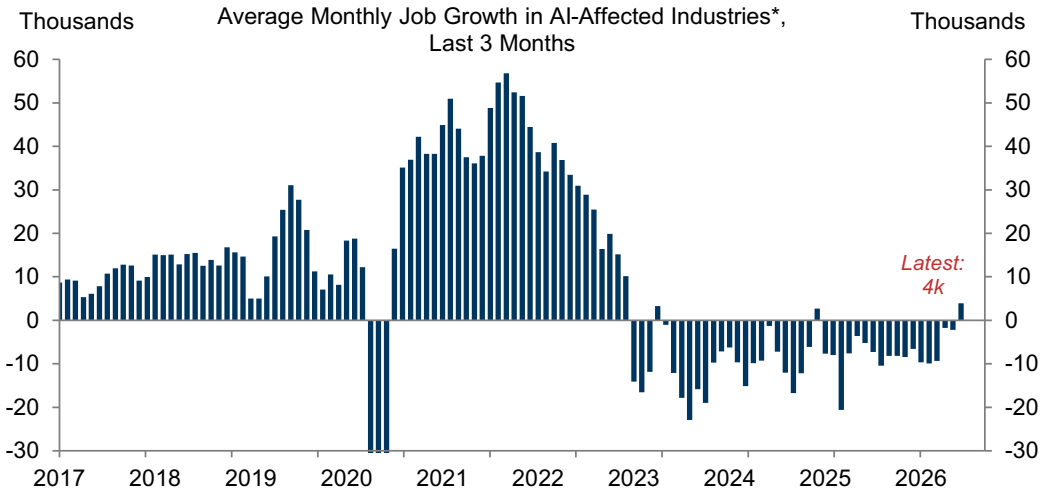
Source: US Bureau of Labor Statistics, IPUMS, Goldman Sachs Global Investment Research

Exhibit 18: Employment Growth Continues to Underperform in Occupations (Specifically Tech) Where Anecdotes Suggest AI Is Having an Impact, but Employment Has Recovered Somewhat in Several Affected Non-Tech Sectors



Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

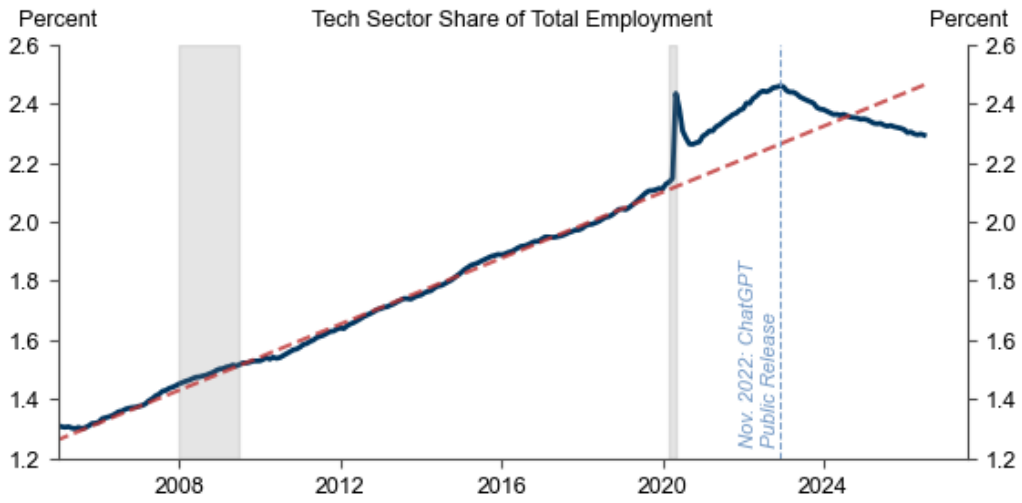
Exhibit 19: Job Growth Drag in AI-Affected Industries Has Reversed For Now



*Management consulting, graphic design, office administration, telephone call centers, computer systems, software publishers, web search, data processing, movie production, broadcasting, publishing, document preparation services.

Source: US Bureau of Labor Statistics, Goldman Sachs Global Investment Research

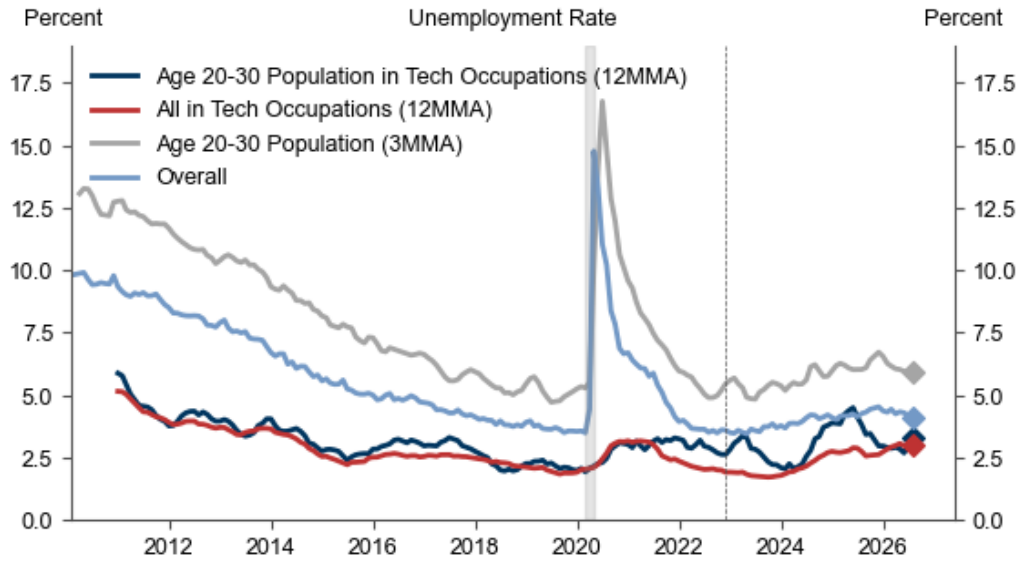
Exhibit 20: Tech's Employment Share Continues to Decline Below Its Long-run Trend



Note: Tech refers to the software publishers, data processing and related, web search and related, and computer systems design subsectors. Red line represents 2007-2019 trend.

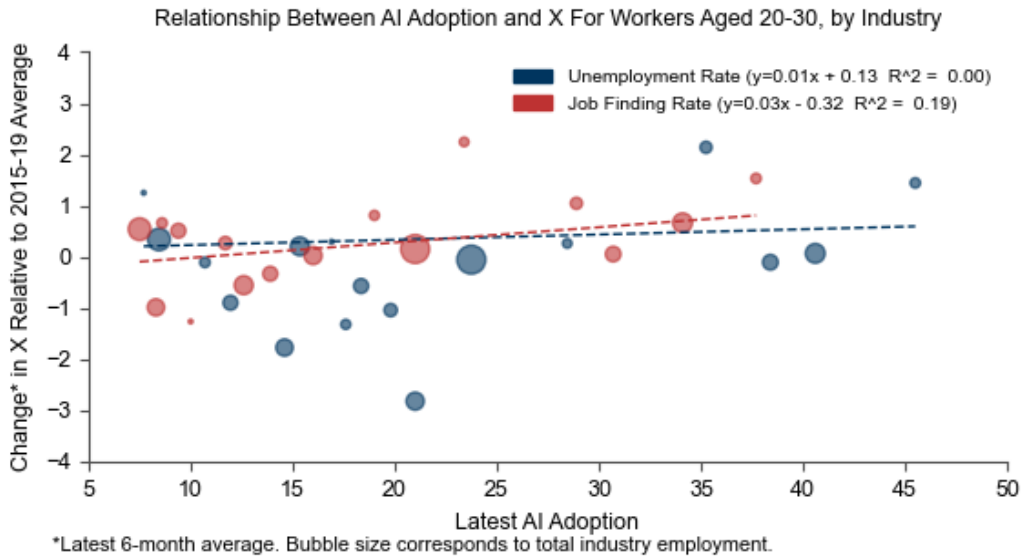
Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 21: Unemployment Among Young Tech Workers Has Fallen Back in Line With Unemployment for All Tech Workers



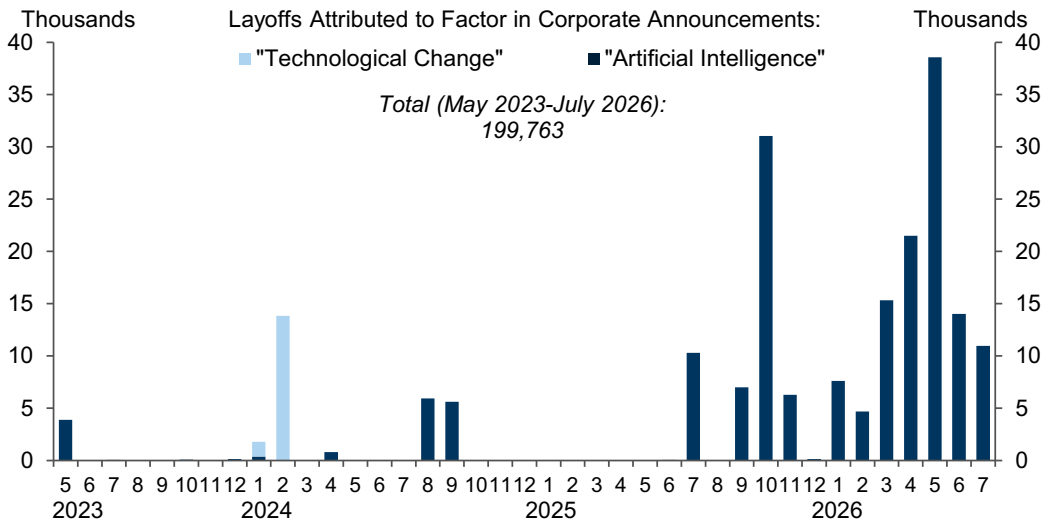
Source: US Bureau of Labor Statistics, IPUMS, Goldman Sachs Global Investment Research

Exhibit 22: We Continue to Observe a Slight Positive Correlation Between AI Adoption and Unemployment Among Younger Workers



Source: IPUMS, US Bureau of Labor Statistics, Census Bureau, Goldman Sachs Global Investment Research

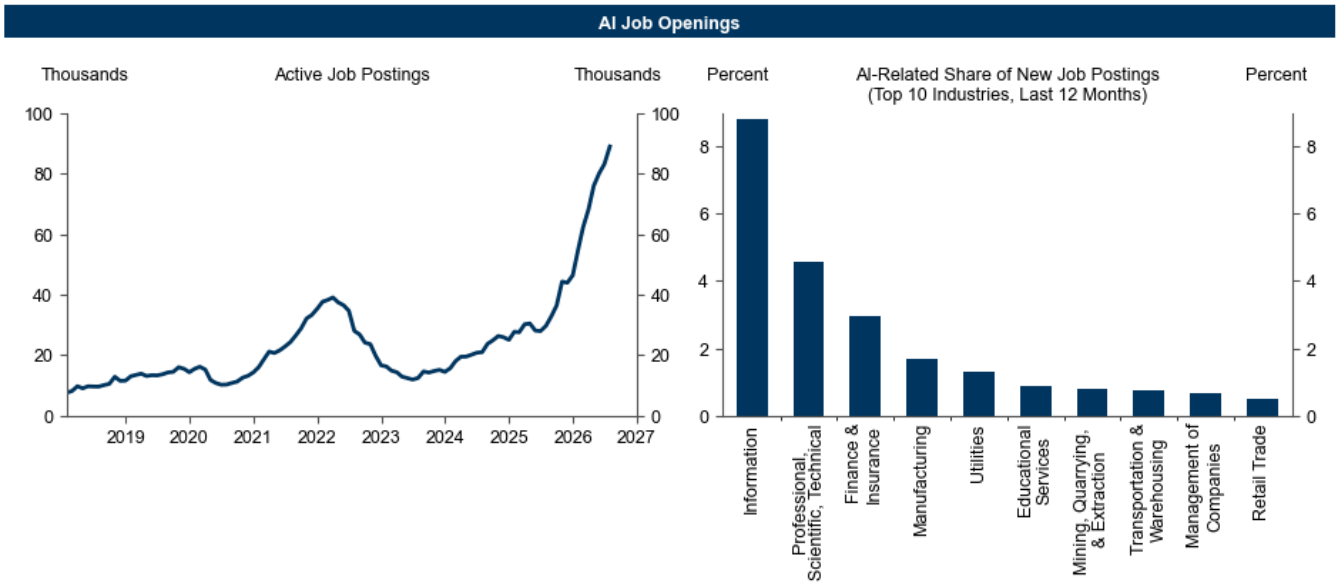
Exhibit 23: AI Was Cited in Layoff Announcements Affecting 11k Employees in July



Source: Challenger, Gray & Christmas, Goldman Sachs Global Investment Research

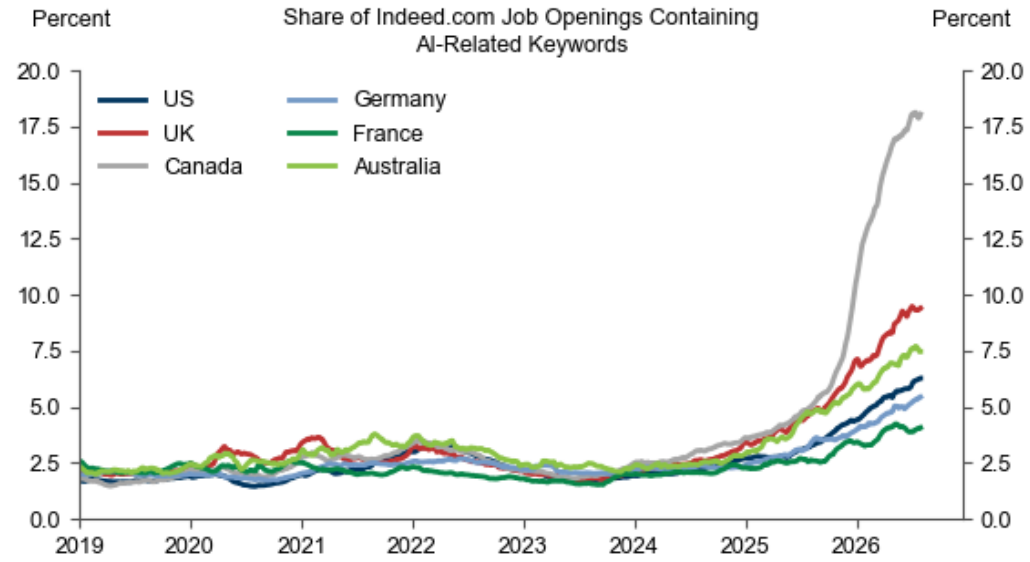
Labor Demand

Exhibit 24: Active AI-Related Job Postings Now Stand at 89k in the US



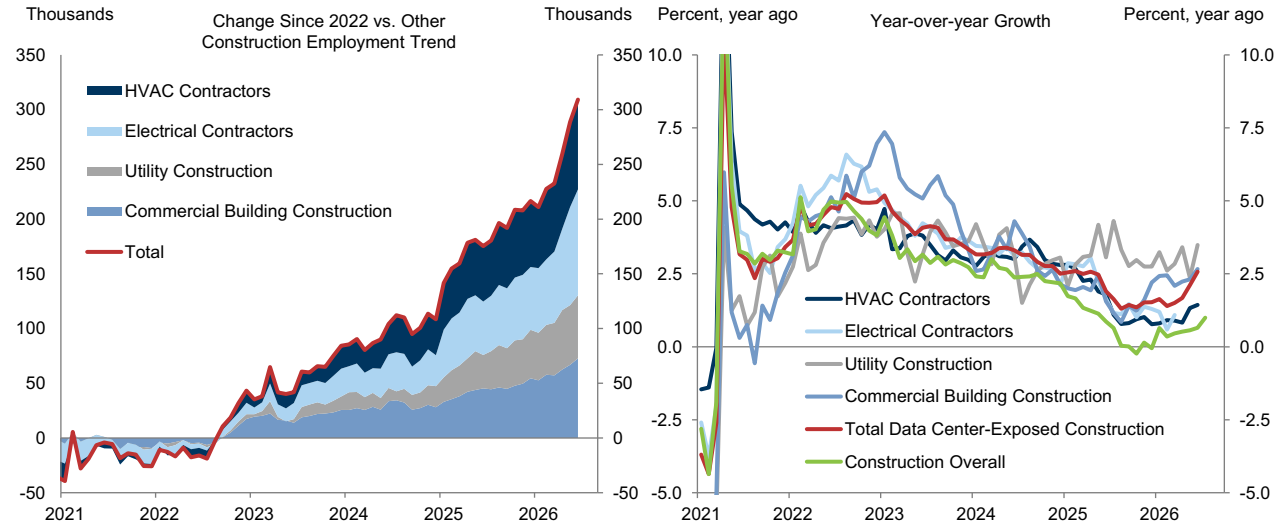
Source: University of Maryland, LinkUp, Goldman Sachs Global Investment Research

Exhibit 25: Share of AI-Related Job Postings Continues to Rise Across All Major DMs



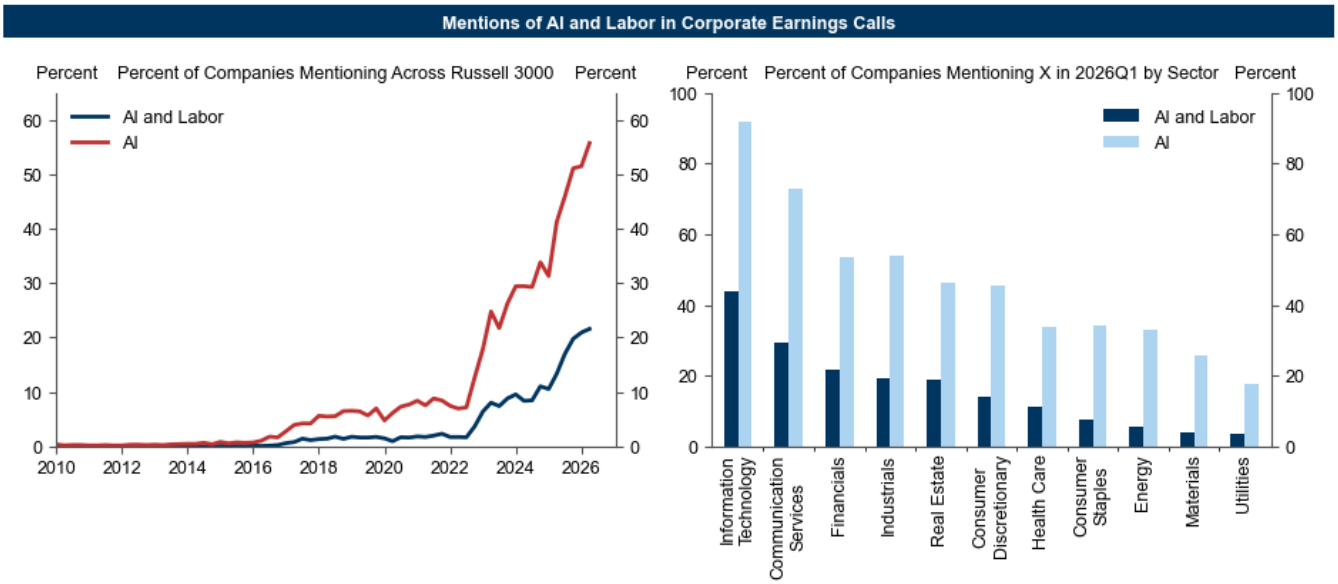
Source: Indeed Hiring Lab, Goldman Sachs Global Investment Research

Exhibit 26: Construction Employment in Data Center-Exposed Categories Has Increased by +309k Since 2022, With a Sharp Acceleration in Recent Months



Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

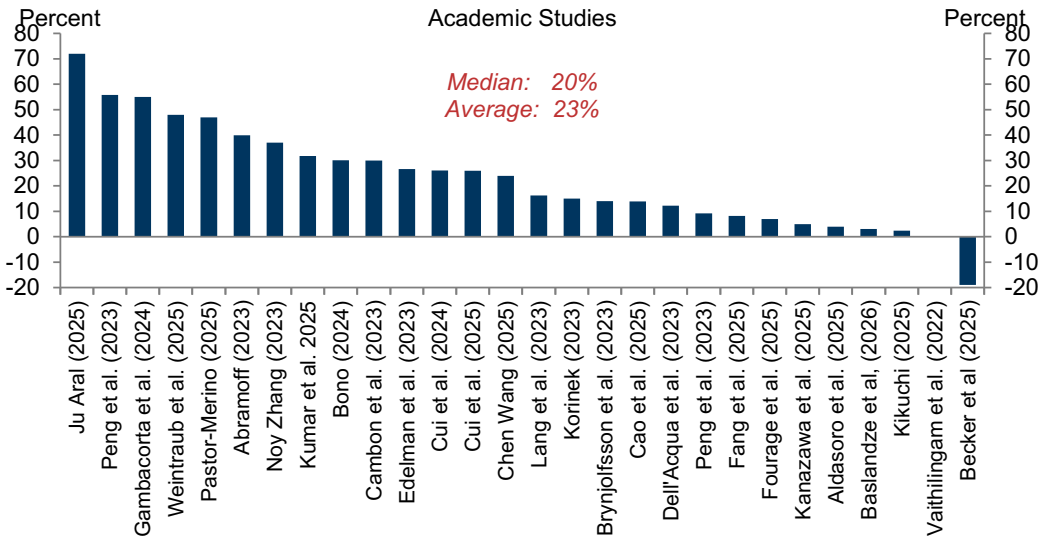
Exhibit 27: 22% of Russell 3000 Companies Mentioned AI and Labor-Related Keywords in 2026Q2 Earnings Calls



Source: GS Data Works, Goldman Sachs Global Investment Research

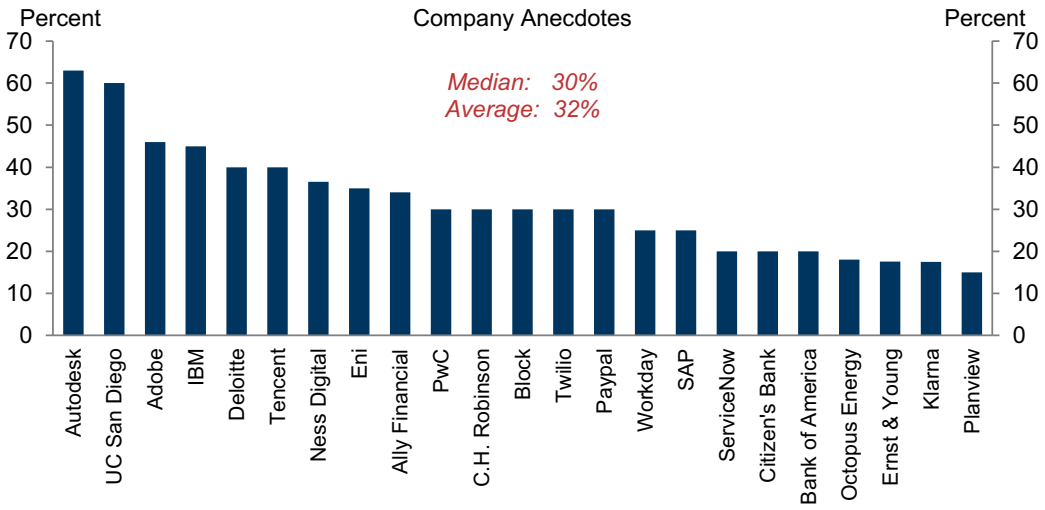
AI's Impact on Productivity

Exhibit 28: Evidence From Academic Studies Suggests AI Adoption Boosts Labor Productivity by Around 23% on Average...



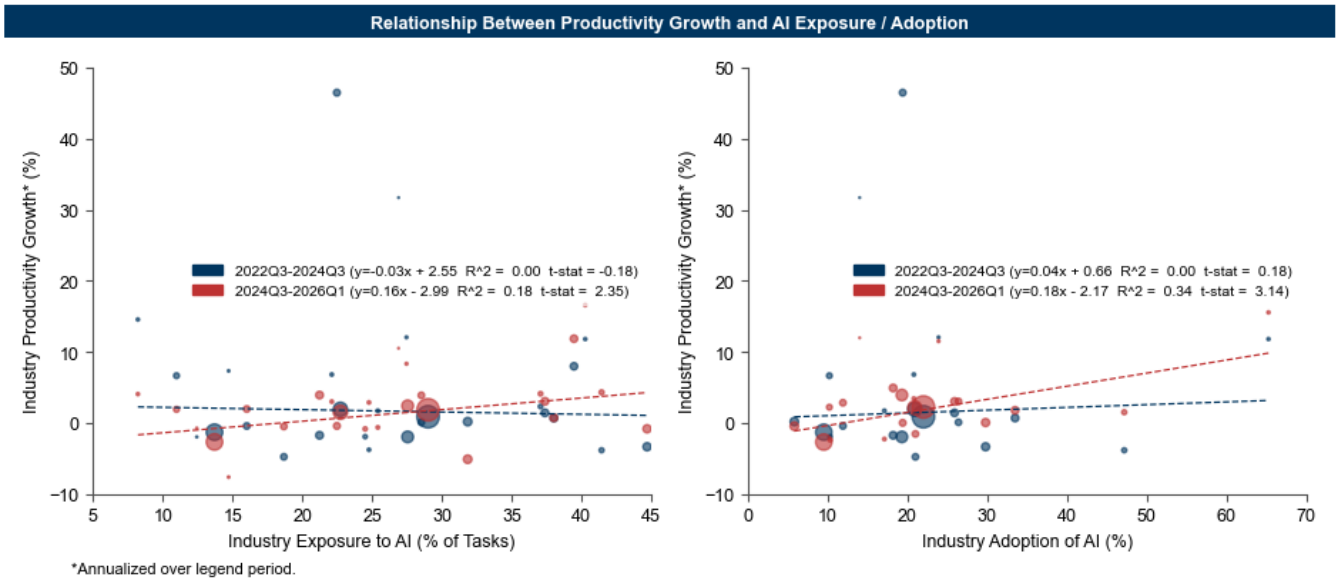
Source: Goldman Sachs Global Investment Research

Exhibit 29: ... And Company Anecdotes Suggest an Average Productivity Boost of 32%



Source: Company data, Data compiled by Goldman Sachs Global Investment Research

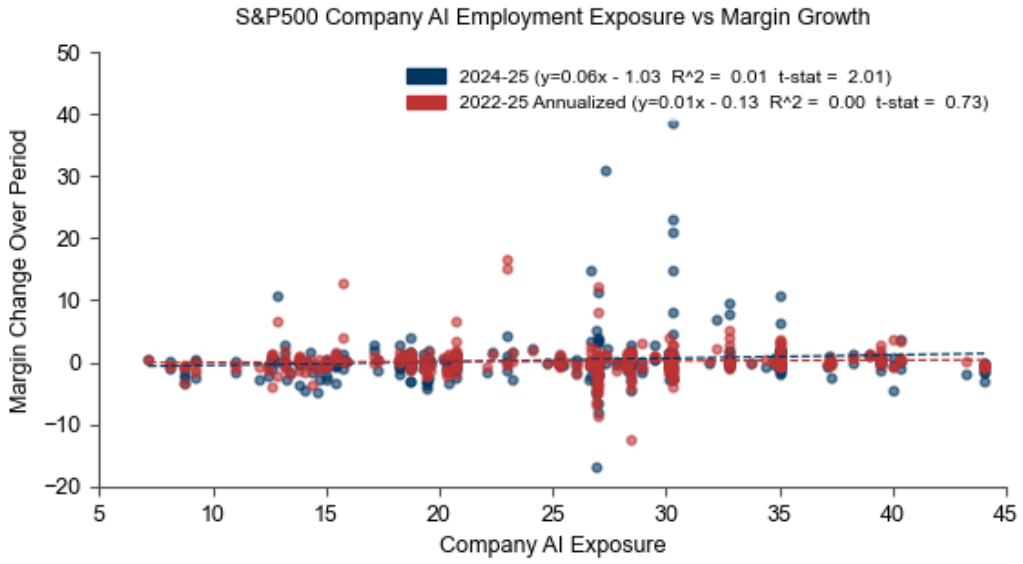
Exhibit 30: Industries with Higher Adoption Rates Continue to Show Slightly Stronger Productivity Growth



Note: Bubble size corresponds to total industry employment.

Source: US Bureau of Labor Statistics, Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 31: No Relationship Between Company Level AI Exposure and Margin Growth



Source: FactSet, Goldman Sachs Global Investment Research

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- US Weekly Kickstart: What Q2 earnings reports signaled about the state of corporate AI adoption, August 14
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